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A Longitudinal Study of Children's Hippocampal Development: Investigating Maternal Physical Activity, Depression, and Education

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ABSTRACT

The developing hippocampus is particularly sensitive to early environmental influences, including during pregnancy. This longitudinal neuroimaging study examined associations between prenatal maternal physical activity and depression, maternal education, and hippocampal development from early childhood to early adolescence. Participants were mothers and their 113 children (59 females; mean age 4.16 ± 1.25 years at first scan) with 510 magnetic resonance imaging scans. Maternal physical activity and depressive symptoms were assessed during the second trimester. Hippocampal diffusion metrics—including fractional anisotropy (FA), mean diffusivity (MD), and radial diffusivity (RD)—as well as volume were measured. Developmental trajectories were analyzed with generalized fractional polynomial mixed models. Results showed significant age-related changes in hippocampal volume and diffusivity, with sex differences in FA development. Bilateral hippocampal volume increased nonlinearly with age, and FA, MD, and RD changed in line with typical brain maturation patterns. However, in contrast to our pre-registered hypotheses, prenatal maternal physical activity was not significantly associated with hippocampal structure. Additionally, in exploratory analyses, we found no significant associations between maternal education or prenatal maternal depression and hippocampal structure. These findings provide a comprehensive characterization of hippocampal development from childhood to adolescence and suggest that prenatal maternal physical activity, depression, and education are not strongly related to hippocampal structure. This work underscores the value of longitudinal neuroimaging and flexible modeling approaches in understanding early brain development.

1 | Introduction

The hippocampus plays a central role in episodic memory, spatial navigation, learning, emotional processing, and neuroendocrine regulation (Insausti and Amaral 2012; White et al. 2024). Anatomically, it is situated within the brain's medial temporal lobe and comprises the cornu ammonis subfields (CA1–CA4), dentate gyrus (DG), and subiculum, which together form a layered, curved structure that protrudes into the temporal horn of the lateral ventricle (Duvernoy et al. 2013; Insausti and Amaral 2012; White et al. 2024). Along its anteroposterior axis, the hippocampus can be subdivided into the head, body, and tail, reflecting regional differences in cytoarchitecture and connectivity (Insausti and Amaral 2012). The hippocampal formation connects to other brain regions through organized intrinsic and extrinsic pathways, receiving input primarily from the entorhinal cortex and projecting via the subiculum and fimbria–fornix (Insausti and Amaral 2012). These pathways enable communication with both cortical and subcortical structures—including the amygdala, mammillary bodies, septal nuclei, hypothalamus, anterior cingulate, and prefrontal cortices—supporting the hippocampus's integrative function across cognitive and affective domains (Insausti and Amaral 2012; White et al. 2024).

In addition to well-documented vulnerability of the hippocampus to neurodegenerative, neuropsychiatric, and neurodevelopmental disorders (Duvernoy et al. 2013; Insausti and Amaral 2012), the hippocampus is also highly sensitive to environmental and lifestyle factors, which can modulate its volume, structural integrity, and functional capacity (Clark et al. 2017; Lupien et al. 2018). Although much foundational work has highlighted the beneficial effects of exercise and physical activity on hippocampal structure and function in adults (Clark et al. 2017; Wilckens et al. 2021), emerging evidence suggests that these neurobiological benefits may begin even earlier in life. Recent animal studies show that prenatal maternal aerobic exercise enhances hippocampal neurogenesis and increases levels of neurotrophic factor—including brain-derived neurotrophic factor (BDNF) and glial cell line-derived neurotrophic factor (GDNF)—in the offspring, as well as improves offspring memory and emotional resilience (Ji et al. 2020; Gomes da Silva et al. 2016; Sabaghi et al. 2019). Yang et al. (2021) conducted a multilevel meta-analysis of 52 rodent studies (4786 animals, 412 effect sizes) and reported a significant 16.5% increase in offspring neurogenesis associated with parental exercise, alongside significant enhancements in cognitive performance, and upregulation of BDNF and vascular endothelial growth factor (VEGF) expression. Although research on the effects of prenatal maternal physical activity and exercise on children's brains remains limited (Marques et al. 2015), a recent scoping review by Valkenborghs, Dent, et al. (2022) identified 14 studies investigating the intergenerational effects of parental physical activity on offspring brain and neurocognitive outcomes. Notably, all observational studies reported positive associations between maternal physical activity—primarily during pregnancy—and improved neurodevelopment. However, only one of four experimental studies found a significant effect, and none directly assessed hippocampal structure. In line with these findings, a systematic review by Niño Cruz et al. (2018) concluded that prenatal leisure-time physical activity was generally associated with enhanced neurodevelopmental outcomes—particularly language skills—in offspring during the first 18 months of life. Most human studies to date have relied on behavioral and cognitive measures, and none have linked prenatal physical activity with hippocampal structure in children. However, recent neuroimaging research has linked prenatal maternal body mass index (BMI) to specific alterations in offspring hippocampal structure. For instance, higher prepregnancy BMI has been associated with higher hippocampal mean diffusivity (MD) in neonates (Rosberg et al. 2025) and lower hippocampal volume in boys (Alves et al. 2020), underscoring the relevance of investigating modifiable maternal behaviors through imaging-based approaches.

Beyond physical activity, prenatal maternal depression has also emerged as a significant influence on child neurodevelopment. Research suggests that maternal depression during pregnancy may alter fetal development through increased maternal cortisol and inflammatory cytokines, disrupted placental function, and epigenetic modifications—mechanisms that may influence hippocampal volume and the child's broader neurodevelopmental trajectories (Center on the Developing Child 2009; Herba et al. 2016; Wu et al. 2020). These disruptions are believed to contribute to lasting changes in brain architecture and stress responsivity (Center on the Developing Child 2009; Herba et al. 2016). Neuroimaging studies examining the effects of prenatal maternal depression on hippocampal structure in children and adolescents remain limited. However, a small body of research in neonates and infants has examined associations between prenatal maternal depressive symptoms and hippocampal volume, with evidence that these associations may vary by infant sex and genetic vulnerability (Acosta et al. 2020; Groenewold et al. 2022; Qiu et al. 2017). Specifically, larger hippocampal volumes have been reported in infants with higher genetic susceptibility (Qiu et al. 2017) and in female infants exposed to antenatal maternal depression (Groenewold et al. 2022). Conversely, Acosta et al. (2020) reported a sex-dependent genetic moderation pattern, consistent with a more negative association between prenatal depressive symptoms and right hippocampal volume among male infants with elevated polygenic risk. More broadly, findings from studies examining maternal lifetime or postnatal depression and hippocampal volume in older children and adolescents are mixed. Chen et al. (2010) reported that healthy girls at high familial risk for depression exhibited lower left hippocampal volume, whereas Hubachek et al. (2021) found both smaller bilateral hippocampal head volumes and an enlarged left hippocampal body in preadolescent offspring of depressed mothers. Conversely, Lupien et al. (2011) observed no significant associations between maternal depression and hippocampal size in children, and Gilliam et al. (2015) identified only indirect effects—namely, altered amygdala–hippocampal volume ratios associated with behavioral outcomes.

Further, accumulating evidence highlights associations between maternal/parental education and hippocampal volume across development. Lower maternal or parental education has been linked to smaller hippocampal volumes in childhood and adolescence (Alex et al. 2024; Noble et al. 2015; Ellwood-Lowe et al. 2018), with Noble et al. (2015) reporting that the association is most pronounced at the lowest levels of parental education.

Together, these studies highlight a critical gap in the literature and underscore the need for objective neuroimaging investigations of associations among prenatal maternal physical activity, depression, maternal education, and the development of the hippocampus. Given that this brain region is characterized by protracted development, high glucocorticoid receptor density, and postnatal neurogenesis, collectively this makes the hippocampus particularly responsive to environmental influences in both negative and positive directions (Clark et al. 2017; Teicher et al. 2003).

This longitudinal study was conceptualized to investigate the effects of prenatal maternal physical activity on the developmental trajectories of hippocampal macrostructural and microstructural metrics. To ensure transparency and rigor in our research process, we preregistered this study on the Open Science Framework (Reynolds et al. 2022) under the title "Prenatal maternal physical activity and hippocampal structure in young children." The preregistration specified a cross-sectional analysis using magnetic resonance imaging (MRI) data acquired closest to 3.5 years of age. With the subsequent availability of a much larger longitudinal dataset spanning ages 2–13 years, we extended the study design to longitudinal analyses of hippocampal trajectories, thereby leveraging the full richness of the dataset. We hypothesized that children born to more physically active mothers would exhibit larger hippocampal volumes and more mature hippocampal microstructure—characterized by higher fractional anisotropy (FA), and lower MD and radial diffusivity (RD)—compared to children born to less active mothers. As a second, exploratory aim, we examined whether prenatal maternal depression and education influenced hippocampal development. We hypothesized that higher prenatal maternal depression would be associated with smaller hippocampal volumes—particularly in the right hemisphere—and greater isotropic diffusion. Conversely, we expected higher maternal education to relate to larger hippocampal volumes and more mature microstructural properties.

2 | Methods

2.1 | Participants

All mothers included in this analysis were non-smokers residing in the Calgary area at the time of data collection. The children in this study are participants in the Calgary Preschool MRI Study (Reynolds et al. 2020) and were originally recruited from the local community and the Alberta Pregnancy Outcomes and Nutrition (APrON) longitudinal study in Calgary, Canada. APrON is a large, ongoing project focused on understanding the early origins of health and disease (Letourneau et al. 2022). Each child was born to a different mother; no siblings or twins were included in the sample. Data on fathers were not available. Initially, 133 participants were scanned, and their MRI data were preprocessed. However, 20 participants were excluded, primarily due to motion artifacts, resulting in a final volumetric analysis sample of 113 typically developing children (54 males, 59 females) aged 1.94–7.36 years (mean = 4.12, SD = 1.23) at first visit. For the hippocampal diffusivity analysis, 10 more participants were excluded, yielding a final sample of 103 participants (46 males, 57 females), aged 1.94–7.36 years (mean = 4.17, SD = 1.22) at first visit. Participants were initially invited for follow-up visits every 6 months for 2 years and annually thereafter, resulting in between 1 and 18 scans per child and a total of 510 datasets (average of 4.51 scans per participant) for the volumetric analysis and between 1 and 12 scans per child, with a total of 429 datasets (an average of 4.17 scans per participant) for the diffusivity analysis (Figure 1). Children had no genetic, physical, neurological, and psychiatric disorders or major intellectual or motor impairments and were born after 35 weeks' gestation. Parental/Guardian written informed consent and child assent (as appropriate) were obtained for each participant. The study was approved by the University of Calgary Conjoint Health Research Ethics Board (REB13-0020).

2.2 | Maternal Characteristics and Prenatal Measures

2.2.1 | Preregistered Analysis: Prenatal Maternal Physical Activity

Habitual physical activity was assessed using the validated Baecke Physical Activity Questionnaire, which measures activity in three domains: work, sport, and leisure (Baecke et al. 1982). Each domain produces scores from 1 to 5, with higher scores indicating greater activity. In the present study, we used the total score (sum of the three domain scores) as a comprehensive measure of physical activity. The work index (eight items) assesses occupational physical demands (e.g., sitting, standing, walking), calculated as work index = ((6 − points for sitting) + sum of other 7 items)/8. Jobs are categorized as low, moderate, or high activity; fewer than eight responses result in missing scores (999). The sport index (four items) evaluates sport frequency, intensity, and effort. A sport score (1–5) is calculated for up to two sports: sport index = sum of 4 parameters/4. Default values apply when fewer sports are reported; missing codes are used if fewer than four items are answered. The leisure index (four items) covers everyday activities (e.g., walking, cycling, and TV watching), calculated as leisure index = ((6 − points for TV watching) + sum of other 3 items)/4, with incomplete responses coded as missing (999). Mothers completed the questionnaire during the first (n = 53), second (n = 120), and third (n = 120) trimesters. In this study, the second trimester was selected as the primary exposure window for analysis. This decision is grounded in neurodevelopmental evidence indicating that the core structural and cellular components of the hippocampus—particularly the CA1–CA3 pyramidal neurons—are largely established by the end of the second trimester (Seress et al. 2001; White et al. 2024). Moreover, the second trimester represents a critical period of hippocampal growth, encompassing peak neurogenesis, dendritic outgrowth, and structural organization (Zhong et al. 2020). As such, physical activity during this trimester is biologically plausible to exert lasting effects on hippocampal volume in offspring. The choice is also supported by adequate sample size (n = 120), allowing for statistically robust analyses. A strong, significant correlation between first- and second-trimester activity levels (r_s = 0.699, p = 8.29 × 10^{-9}, 95% CI: 0.520–0.819) indicates that physical activity was generally stable across these periods. In addition to the preregistered analyses using the total Baecke Physical Activity score, we conducted supplementary analyses focusing on the sport subscale alone to provide a more targeted mea-

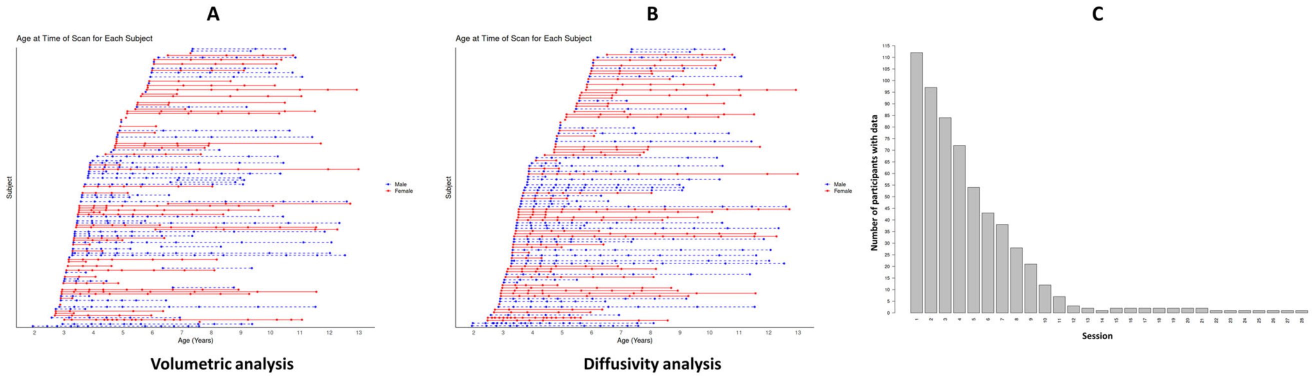

FIGURE 1 | Age at the time of each MRI scan for all participants included in the longitudinal sample. (A) Scan sessions included in the volumetric analysis. (B) Scan sessions included in the diffusivity analysis. (C) Number of participants with data at each scan session (visit number).

sure of leisure-time activity. This decision was motivated by evidence suggesting that occupational physical activity may not confer the same health benefits as leisure-time physical activity (Holtermann et al. 2018; de Vries and Bakker 2022).

2.2.2 | Exploratory Analysis: Prenatal Maternal Education and Depression

Self-reported maternal education was categorized into four groups: (1) some/finished high school, (2) some/finished college, (3) undergraduate, and (4) some postgraduate. Education data were missing for four participants. Family income ranged from under $25,000 to over $175,000, with a median income range of $125,000–$149,999.

Maternal depressive symptoms during the second trimester of pregnancy were assessed using the Edinburgh Postnatal Depression Scale (EPDS)—a validated and widely used 10-item self-report questionnaire, originally developed for postpartum depression but since adapted for antenatal screening (Bergink et al. 2011). The EPDS evaluates symptoms over the past 7 days and generates a total score ranging from 0 to 30, with higher scores indicating greater depressive symptomatology. The EPDS demonstrates high internal consistency, test–retest reliability, and concurrent validity with anxiety and somatization symptoms during pregnancy (Bergink et al. 2011).

To assess potential interrelationships among maternal predictors, we examined pairwise correlations between maternal education, EPDS, and second-trimester maternal physical activity (Baecke total score). Distributional assumptions were evaluated using the Shapiro–Wilk test. Given the ordinal nature of maternal education and non-normal distributions of EPDS scores, Spearman's rank correlation coefficients (r_s) were calculated using pairwise complete observations.

2.3 | MRI Acquisition and Analysis

Participants were scanned using a research-dedicated 3T GE MR750w MRI scanner equipped with a 32-channel head coil. Whole-brain T1-weighted anatomical images were acquired with an FSPGR BRAVO sequence ($0.9 \times 0.9 \times 0.9$ mm^3 resolution; 210 axial slices; TR = 8.23 ms; TE = 3.76 ms; flip angle = 12°; matrix size = 512×512; inversion time = 540 ms; total imaging time = 4 min 26 s). All T1-weighted images were inspected for motion artifacts while the participant was still in the scanner, allowing scans to be repeated if necessary. All T1-weighted scans included in the final analysis passed quality checks for motion and scanner-related artifacts.

Diffusion-weighted imaging (DWI) data were acquired using a single-shot spin echo echo-planar imaging sequence (TR = 6750 ms, TE = 97 ms, $1.6 \times 1.6 \times 2.2$ mm^3 resolution [resampled on scanner to $0.78 \times 0.78 \times 2.2$ mm^3], FOV = 20.0, 30 gradient encoding directions at b = 750 s/mm^2, and five interleaved volumes at $b = 0$ s/mm^2 for a total acquisition time of 4:03 min:s).

T1-weighted images corrected for intensity bias using the N4 algorithm (Cox 1996; Cox and Hyde 1997) were resampled to an isotropic voxel size of 1 mm^3 to enable multi-atlas segmentation integrated with cortical surface reconstruction via implicit surface evolution (MaCRUISE; Huo, Carass, et al. 2016; Huo, Plassard, et al. 2016; Huo et al. 2018). MaCRUISE is an open-source pipeline that integrates brain segmentation and cortical surface reconstruction into a topologically consistent framework (Huo, Plassard, et al. 2016). Unlike pipelines such as FreeSurfer, MaCRUISE performs segmentation within a unified framework, enhancing anatomical accuracy and robustness for neuroimaging studies involving neurodevelopment (Long et al. 2024) and aging (Huo, Plassard, et al. 2016). Skull- and dura-stripped MRI scans underwent both multi-atlas segmentation of 132 regions (Asman and Landman 2012, 2013; Bermudez et al. 2020; Klein et al. 2010) and TOADS fuzzy membership segmentation (Bazin and Pham 2008). These outputs were fused to produce a comprehensive cerebrum segmentation encompassing both gray and white matter. The resulting tissue labels defined the outer and inner cortical surfaces, which were then used to refine the initial segmentation, ensuring alignment between surface geometry and volumetric labels. Longitudinal registration was applied to prevent biologically implausible changes that can result from the accumulation of minor segmentation errors introduced by the MaCRUISE pipeline, which might distort individual longitudinal trajectories (Roeske et al. 2025). For each participant, the T1-weighted image closest to the median scan was registered to all other time points using rigid, affine, and nonlinear transformations (Ourselin et al., 2001; 2002). The transformation fields generated from this process were then applied to the MaCRUISE segmentation of the median scan, warping it into the native space of each earlier and later time point. The longitudinally registered T1-weighted images were reviewed for the accuracy of hippocampal segmentation, and minor corrections were applied as necessary to ensure precise boundary delineation, as described by Long et al. (2024). MRI datasets with poor segmentation were excluded from the analysis. The total intracranial volume (ICV) was computed by summing the volumes of all segmented regions. Hippocampal volumes for the left and right hemispheres were measured separately for each scan.

DWI data were preprocessed using the open-source MRtrix3 software tool (Tournier et al. 2019). Preprocessing steps included correction for Gibbs's ringing (Kellner et al. 2016), eddy currents, and motion-related artifacts (Andersson and Sotiropoulos 2016). Identification and replacement of outlier slices was done using a Gaussian prediction model during preprocessing (Andersson et al. 2016). The preprocessed images were then resampled to an isotropic voxel size of 1 mm^3 to match the dimensions of the MaCRUISE T1w segmentations. Following preprocessing, the diffusion tensor model was applied to estimate the principal direction of diffusion within white matter voxels, and scalar maps of FA, MD, and RD were computed.

All scans were then quality checked using a combination of quantitative and visual approaches. Quantitative assessment was performed using quality control reports generated via FSL EDDY during preprocessing (Bastiani et al. 2019). Good quality scans were identified if they contained <10% of outlier slices and an average relative motion of <0.7 mm across volumes. Visual assessment involved inspecting directionally encoded color (DEC) maps to evaluate the accuracy of the tensor-model in estimating the orientation of major white matter tracts (i.e., tracts oriented along the inferior–superior axis should be colored blue, anterior–posterior tracts green, and left–right tracts red).

Hippocampal masks for the left and right hemispheres were derived from the MaCRUISE segmentations. For each participant, FSL FLIRT was used to affinely register the MaCRUISE hippocampal segmentations from structural space to diffusion space using the participant's FA map as the reference image. Nearest-neighbor interpolation was used to preserve the discrete nature of atlas labels and anatomical boundaries during transformation. Registration quality was assessed by overlaying the transformed segmentations on the FA map. The transformed left and right hippocampal labels were then extracted and binarized to create regions of interest, which were used to compute mean hippocampal values from the corresponding diffusion maps (FA, MD, RD).

2.4 | Statistical Analysis

All descriptive and inferential statistics were carried out using RStudio (Posit Team 2025), an integrated development environment for R. Normality of the distributions for prenatal maternal physical activity and depression was assessed using the Shapiro–Wilk test. On the basis of the results, appropriate descriptive statistics were provided for each variable. For normally distributed continuous variables, means and standard deviations are reported; for non-normally distributed and ordinal variables, medians, interquartile ranges, and percentages are reported.

Given the longitudinal design of this study, statistical analyses accounted for the dependency of repeated observations within individuals. Linear mixed-effects models (LMMs) are commonly used in developmental neuroimaging due to their ability to model intra-individual change while handling unbalanced data structures resulting from missing observations and irregular time intervals (Vijayakumar et al. 2018). Although LMMs typically model developmental trajectories using linear, quadratic, or cubic terms, such polynomials can impose restrictive shapes and may not adequately capture the complexity of neurodevelopment. For instance, white matter and hippocampal development often follow nonlinear trajectories during childhood and adolescence (Lebel et al. 2019). Fjell et al. (2010) demonstrated that quadratic models, commonly used to model parabolic-like age trajectories in brain structures, can be heavily affected by the age range of participants included in the study. As a result, Fjell et al. (2010) recommended using nonparametric smoothing splines, which offer greater flexibility in capturing complex patterns. However, because smoothing splines are local nonparametric methods, they are generally not ideal for making statistical inferences.

To address these limitations, we utilized the generalized fractional polynomial mixed model (GFPMM; Ryoo et al. 2017), which offers greater flexibility in modeling nonlinear patterns while maintaining suitability for inferential analysis. Unlike standard polynomial models limited to integer powers, GFPMMs allow for both integer and non-integer transformations of time (e.g., age), offering a broader range of possible models. In this study, we evaluated fractional polynomial degrees 1 (FP1) and 2 (FP2), with powers selected from the predefined set $\{-2, -1, -0.5, 0, 0.5, 1, 2, 3\}$, where 0 represents a logarithmic transformation. For FP1, the model takes the form $\beta_0 + \beta_1(t^{\mathrm{P}})$, whereas for FP2, if $P_1 \neq P_2$, the model is expressed as $\beta_0 + \beta_1\ (t^{\mathrm{P1}}) + \beta_2\ (t^{\mathrm{P2}})$; however, if $\mathrm{P1} = \mathrm{P2}$, the model becomes $\beta_0 + \beta_1\ (t^{\mathrm{P1}}) + \beta_2\ (t^{\mathrm{P1}}) \times (\log t)$. A total of 8 FP1 models and 36 FP2 models are assessed based on these power combinations (Royston and Sauerbrei 2008; Aghamohammadi-Sereshki et al. 2022). This flexibility makes GFPMMs particularly useful for modeling developmental trajectories where growth patterns are asymmetric or exhibit nonlinear trends. Additionally, GFPMMs account for individual variability with subject-specific random intercepts and slopes, making them well-suited for modeling repeated measures data with irregular time intervals and unbalanced datasets. This approach has been shown to outperform conventional polynomial models in terms of model fit and parsimony, particularly in contexts where nonlinear trajectories are expected (Ryoo et al. 2017). The GFPMM formulation used in this study follows Ryoo et al. (2017) and can be expressed as

$$\eta_{ij} = \beta_0 + \sum_{b=1}^{p} \beta_b f_b\left(t_{ij}\right) + \sum_{e=1}^{l} \gamma_e X_{ej} + b_{0i} + \sum_{g=1}^{q} b_{ig} f_g\left(t_{ij}\right)$$
$$= X_i \beta + Z_i b$$

where η_{ij} is the hippocampal structural indices for the jth measurement of the ith individual; $f_b(t_{ij})$ and $f_g(t_{ij})$ represent fractional polynomial transformations of the time variable for the fixed effects and the random effects, respectively; X_{ej} denotes static covariates included in the model; and b_{0i} and b_{ig} are random effects accounting for between-subject variability.

Data were cleaned and preprocessed using the *tidyverse* suite in R (Wickham et al. 2019). Missing data in baseline maternal variables (maternal education, household income at the first visit, second-trimester maternal physical activity [Baecke total], and second-trimester depressive symptoms [EPDS]) were handled using multiple imputation by chained equations (MICE) implemented with the mice package in R (van Buuren and Groothuis-Oudshoorn 2011), under a missing-at-random assumption, to minimize bias and loss of precision associated with complete-case analysis (Austin et al. 2021; Jakobsen et al. 2017). Predictive mean matching was used for all imputed variables to preserve plausible values consistent with the observed distributions, and 30 imputed datasets were generated to reflect imputation uncertainty. Imputation diagnostics (missing-data patterns and graphical comparisons of observed vs. imputed distributions) indicated acceptable performance. Subsequent longitudinal mixed-effects analyses were conducted in each imputed dataset, and estimates were combined across imputations using Rubin's rules (Rubin 1987). All models were fitted using the *lme4* and *lmerTest* packages in R (Bates et al. 2015; Kuznetsova et al. 2017), with random intercepts and, where supported by the data, random slopes for age. To evaluate the developmental trajectory of the hippocampus across early childhood using GFPMM analysis, hippocampal structural indices were modeled as dependent variables in separate regression models. Fixed effects included age, sex, an age × sex interaction term, income, maternal education, prenatal maternal physical activity (total score or sport subscale), and depression. ICV was included as a covariate only in models assessing hippocampal volume, but not in models of diffusivity parameters. Age × sex interaction was included ($p < 0.05$) but removed from the model if not significant.

In line with our OSF preregistration, which specified a cross-sectional evaluation of associations between prenatal maternal factors and hippocampal structure using a single time point per child, we conducted cross-sectional regression analyses using data from the first MRI visit (baseline) for each participant. Separate models were fit for left and right hippocampal volume and for DTI-derived hippocampal metrics (FA, MD, and RD). In all cross-sectional models, hippocampal measures at the first visit served as the dependent variables. Independent variables included age at first visit, sex, an age × sex interaction term, income at the first visit, maternal education, maternal physical activity, and depression. For volumetric analyses, ICV was included as a covariate to account for individual differences in head size, whereas ICV was not included in models of diffusivity metrics. Age was modeled flexibly using multivariable fractional polynomial regression to allow for potential nonlinear effects (Royston and Sauerbrei 2008; Aghamohammadi-Sereshki et al. 2022). Age × sex interaction was included ($p < 0.05$) but removed from the model if not significant. All models were estimated using Gaussian regression, and results are reported as standardized β with 95% confidence intervals and p values.

To determine the optimal age model, we used the *mfp* package (Ambler and Benner 2024) and the closed testing procedure (Royston and Sauerbrei 2008; Aghamohammadi-Sereshki et al. 2022), which sequentially compares nested models (linear, FP1, FP2) using likelihood ratio tests (LRTs). Improvements in model fit were evaluated using chi-square (χ^2) statistics and a significance threshold of $p < 0.05$. This rigorous selection procedure ensured that the final model for age was both parsimonious and statistically justified. To assess the contribution of additional predictors to hippocampal structural indices, other covariates were included as fixed effects in the final age models. Significance was evaluated using regression coefficients and t-tests with Satterthwaite's approximation, allowing us to estimate the unique effect of each covariate while accounting for age-related change and repeated measures. $p < 0.05$ was used for the selection of all covariates. To control the familywise Type I error rate, we applied a Bonferroni correction to each regression coefficient's p value, setting the significance threshold at α/k—where k is the number of predictors in each model (Mundfrom et al. 2006). This procedure maintains the overall false-positive rate at or below the nominal α level. In Section 3, p values that remain significant after Bonferroni correction are denoted as *p_Bonf*.

3 | Results

3.1 | Descriptive Statistics

The Shapiro–Wilk test indicated that prenatal maternal physical activity did not significantly deviate from normality, $W = 0.983$, $p = 0.239$, whereas depression scores were skewed toward lower values ($W = 0.904$, $p < 0.001$). Most mothers reported "undergraduate" or higher levels of education, and the largest proportion of households reported incomes over $175,000. Details on sample size, missing data, central tendency, and category breakdowns are provided in Table 1.

Spearman correlations among maternal predictors were small in magnitude, with a weak positive association between maternal education and physical activity ($r_s = 0.21$, $p = 0.037$), and no statistically significant associations between education and EPDS ($r_s = -0.11$, $p = 0.26$) or between EPDS and physical activity ($r_s = -0.19$, $p = 0.06$).

3.2 | Longitudinal Analysis

3.2.1 | Hippocampal Volume

Prenatal maternal physical activity, prenatal maternal depression, maternal education, and household income were not significant predictors of right or left hippocampal volume (all *p_Bonf* > 0.20). Hippocampal volume increased nonlinearly with age in both hemispheres (right: age^{-2}, *p_Bonf* = 3.54e − 9; left: age$^{-0.5}$, *p_Bonf* = 4.70e − 13), plateauing over time (Figure 2). ICV was a strong positive predictor in both hemispheres (right: *p_Bonf* = 1.80e − 13; left: *p_Bonf* = 8.4e − 15). No significant main effect of sex or age × sex interaction was observed for hippocampal volume in either hemisphere (all $ps > 0.20$). See Table 2 for detailed estimates, standardized coefficients, and confidence intervals.

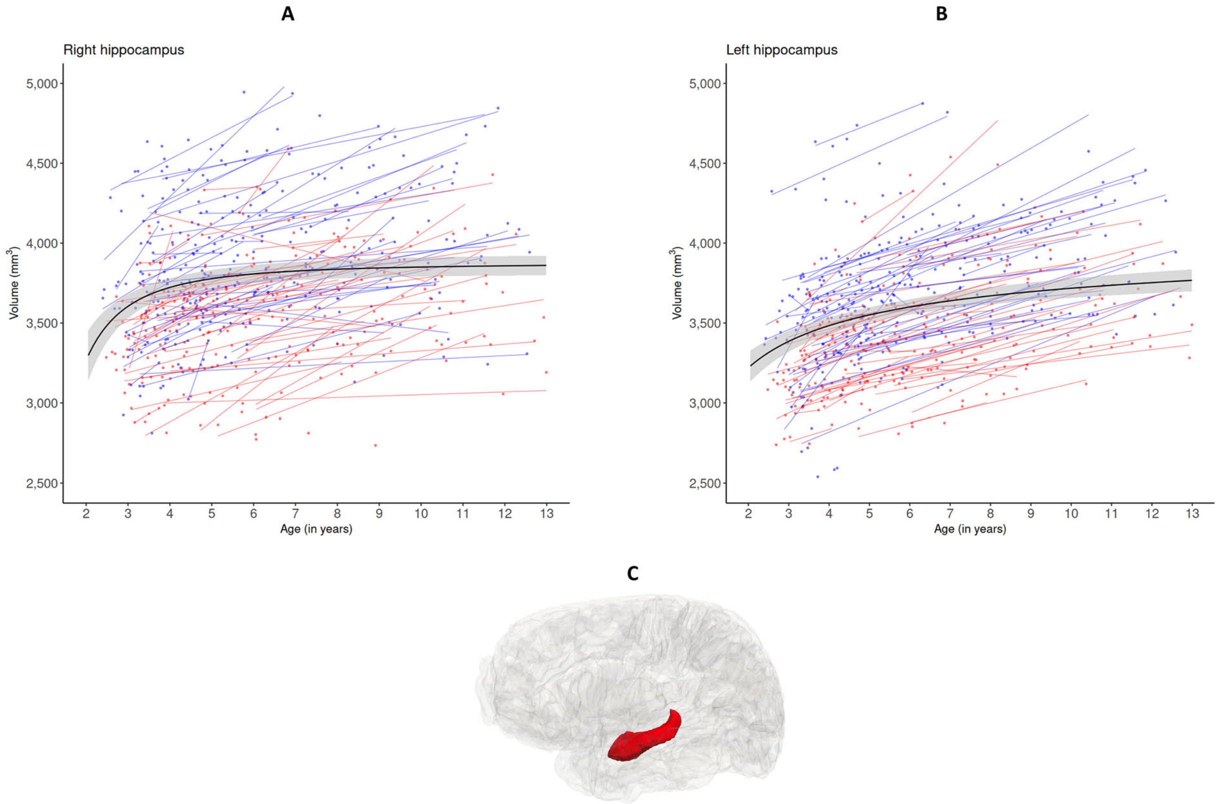

FIGURE 2 | Regression plots illustrating developmental changes in hippocampal volume: (A) right hippocampus and (B) left hippocampus. The black line represents the group-level model fit. Thin blue and red lines indicate individual linear fits for males and females, respectively. Each blue and red dot represents a data point from a male or female participant. (C) Lateral view of a 3D glass brain visualization highlighting the hippocampus.

3.2.2 | Hippocampal Diffusivity

Across all models, prenatal maternal physical activity, prenatal maternal depression, maternal education, and household income were not significant predictors of hippocampal FA, MD, or RD in either hemisphere ($ps \geq 0.10$). When the sport subscale of the Baecke questionnaire was analyzed in the regression models, the results were unchanged: No significant associations were observed between prenatal maternal sport-related physical activity and hippocampal volume or diffusivity indices, and findings for other regressors were unaffected.

For FA, a significant age × sex interaction was observed in both hemispheres (both $p\_Bonf < 0.006$), indicating different developmental trajectories in males and females. In both sexes, right hippocampal FA initially decreased with age, followed by a plateau and gradual increase—modeled as $\text{age}^{-0.5}$ and $\text{age}^{0.5}$ in males (both $p\_Bonf < 0.0003$) and age^{-2} and age^{-1} in females (*both* $p\_Bonf < 0.0007$). Both the decline and subsequent rise in FA were steeper in females, suggesting more dynamic white matter maturation (Figure 3). In both sexes, left hippocampal FA initially decreased with age, followed by a plateau—modeled as Age^{-2} in males ($p\_Bonf < 0.0006$) and as a second-degree fractional polynomial (FP2; $\text{age}^{-0.5}$ and $\text{age}^{-0.5}$) in females (Figure 3). Although the FP2 model provided the best statistical fit for females, indicating a subsequent increase after the plateau, this pattern did not retain statistical significance after Bonferroni correction for multiple comparisons (*p_Bonf*: 0.225 and *p_Bonf*: 0.058, respectively). Thus, although females showed a potential nonlinear increase after the plateau, this result must be interpreted with caution, as the adjusted significance suggests uncertainty about whether this observed effect is genuine or due to chance. No similar increase was observed in males. Sex significantly predicted FA only in the left hemisphere (*p_Bonf* for the right: 0.063; and the left <0.0007), with females showing lower left-hemisphere FA.

MD decreased rapidly with age in both hemispheres (*p_Bonf* for the right (age^{-2}) = 1.30e − 15; and the left (age^{-1}) = 1.30e − 15), plateauing over time, with a steeper decline in the right hemisphere suggesting faster maturation (Figure 4). No significant main effect of sex or age × sex interaction was found for MD (all $p\_Bonf > 0.053$).

RD showed a decrease in both hemispheres (*p_Bonf* for the right (age^{-1}) < 1.30e − 15 and for the left ($\text{age}^{-0.5}$) = 1.15e − 14), plateauing over time, with a steeper decline in the right hemisphere suggesting accelerated maturation (Figure 5). No significant main effect of sex or age × sex interaction was found for RD (all $p\_Bonf > 0.16$).

3.3 | Cross-Sectional Analysis

3.3.1 | Hippocampal Volume

In cross-sectional analyses, prenatal maternal physical activity, prenatal maternal depression, maternal education, and house-

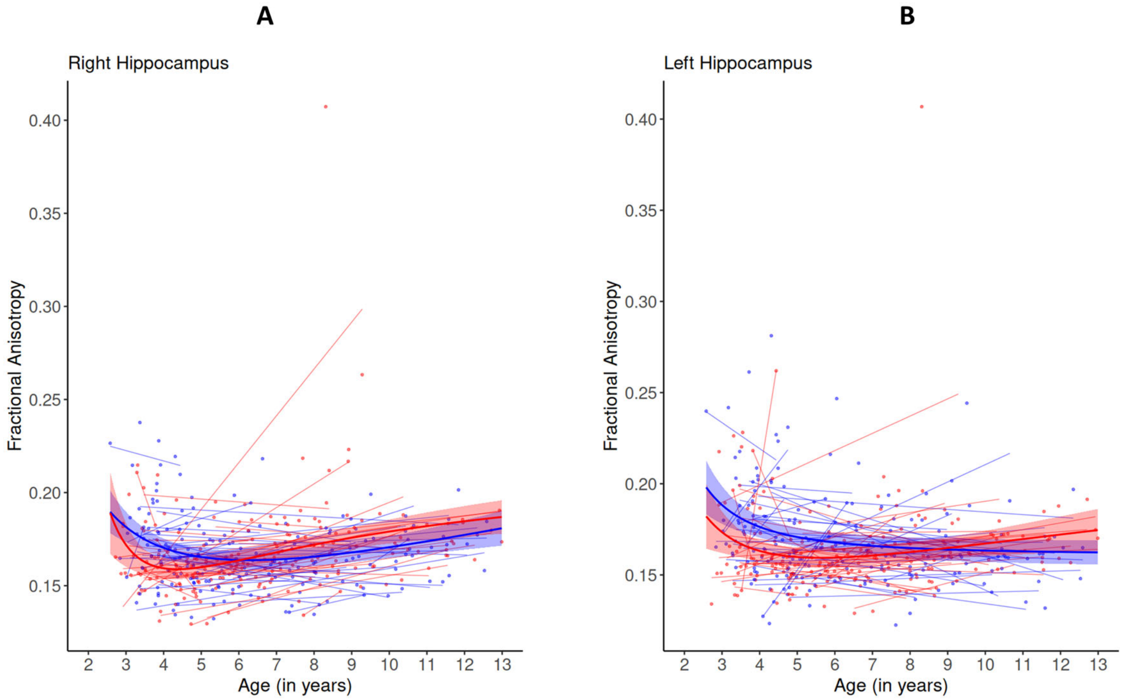

FIGURE 3 | Regression plots illustrating developmental changes in hippocampal FA: (A) right hippocampus and (B) left hippocampus. Thick blue and red lines represent the group-level model fit for males and females, respectively. Thin blue and red lines indicate individual linear fits for males and females, respectively. Each blue and red dot represents a data point from a male or female participant.

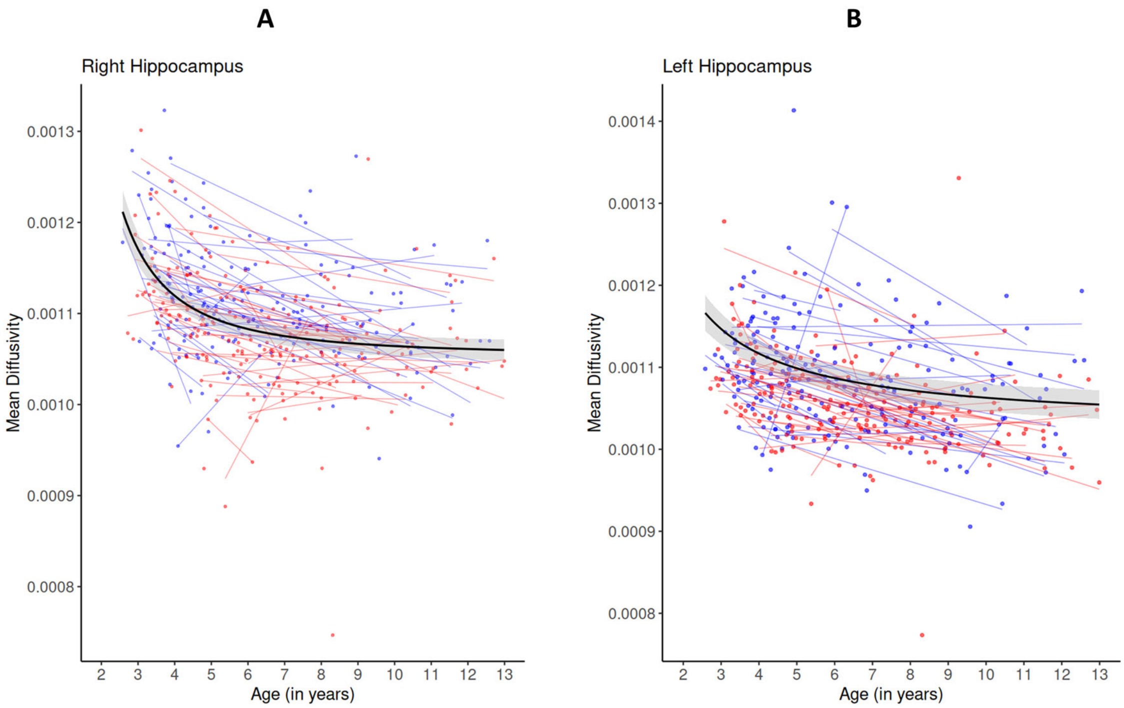

FIGURE 4 | Regression plots illustrating developmental changes in hippocampal MD (mm^2/s): (A) right hippocampus and (B) left hippocampus. The black line represents the group-level model fit. Thin blue and red lines indicate individual linear fits for males and females, respectively. Each blue and red dot represents a data point from a male or female participant.

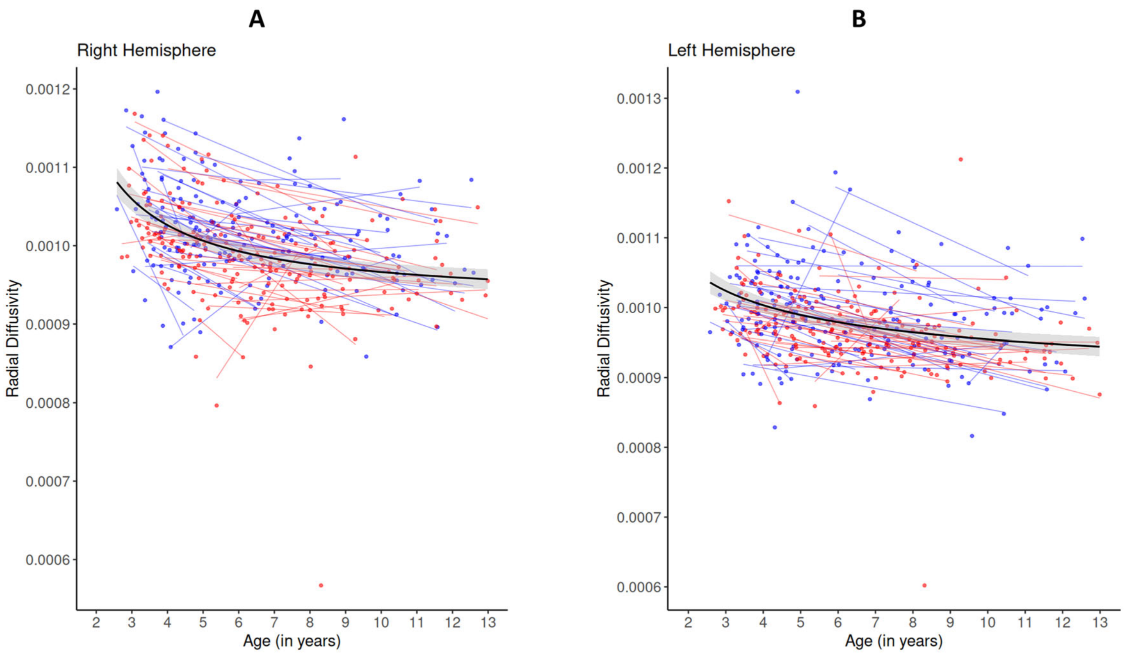

FIGURE 5 | Regression plots illustrating developmental changes in hippocampal RD (mm^2/s): (A) right hippocampus and (B) left hippocampus. The black line represents the group-level model fit. Thin blue and red lines indicate individual linear fits for males and females, respectively. Each blue and red dot represents a data point from a male or female participant.

hold income were not significant predictors of hippocampal volume after Bonferroni correction in either hemisphere (all $p\_Bonf \geq 0.497$). Age was also not significant in either hemisphere after correction (both $p\_Bonf > 0.07$). No significant main effect of sex or age × sex interaction was found for volume in either hemisphere. See Table 3 for full standardized coefficients and confidence interval.

3.3.2 | Hippocampal Diffusivity

Cross-sectional analyses of hippocampal diffusivity metrics (FA, MD, and RD) were conducted with Bonferroni correction for multiple comparisons. Overall, limited evidence for cross-sectional associations was observed after correction. Prenatal maternal physical activity, prenatal maternal depression, maternal education, and household income were not significantly associated with hippocampal FA, MD, or RD in either hemisphere after correction for multiple comparisons (all $p\_Bonf > 0.11$).

Among the diffusivity measures, MD in the right hippocampus showed a significant association with age (p_Bonf: 0.018) that remained statistically significant after Bonferroni correction, indicating lower MD values with increasing age. In contrast, no other age-related associations for FA or RD in either hemisphere retained statistical significance following correction. Associations between age and FA or RD, although present at the uncorrected level in some models, did not survive adjustment for multiple comparisons (all $p\_Bonf > 0.07$), suggesting that these effects should be interpreted cautiously.

Finally, except for right hippocampal MD, which was significantly higher in males than females (p_Bonf: 0.036), there were no significant main effects of sex and no age × sex interactions across DTI metrics after multiple-comparisons correction in either hemisphere (all $p\_Bonf > 0.07$).

Complete standardized coefficients, confidence intervals, and corrected significance levels for all cross-sectional diffusivity models are reported in Table 3.

4 | Discussion

Our study found that prenatal maternal physical activity, prenatal maternal depression, maternal education, and household income were not significantly associated with hippocampal volume or microstructure. In contrast, the longitudinal analysis showed that hippocampal volume increased nonlinearly with age in both hemispheres, with a steeper increase in the left hippocampus. Diffusivity metrics also showed age-related developmental changes, and sex differences in developmental trajectories were observed for hippocampal FA in both hemispheres.

In older adolescents and adults, physical activity is associated with increased anterior hippocampal volume (Erickson et al. 2011), improved functional connectivity between the hippocampus and other brain regions (Burdette et al. 2010; Voss et al. 2010), and improved hippocampal metabolism (Valkenborghs, Hillman, et al. 2022). Notably, a meta-analysis of 23 intervention studies reported that moderate, prolonged exercise has positive effects on hippocampal volume in older adults (Wilckens et al. 2021). A growing body of human research suggests that maternal physical activity during pregnancy may similarly influence early neurodevelopment. Several cohort and randomized controlled trials have reported modest positive effects on the cognitive, language, and behavioral outcomes of infants, children, and adolescents (H. Marques et al. 2015; Leão et al. 2022; Moyer

TABLE 1 | Demographic and descriptive statistics of study variables.

Variable		N^{a}	Central tendency	Range	Notes/Categories
Age (first visit)	Volumetric analysis	113	Mean = 4.12, SD = 1.23	1.94–7.36 years	54 males, 59 females
	Diffusivity analysis	103	Mean = 4.17, SD = 1.22	1.94–7.36 years	46 males, 57 females
Prenatal maternal physical activity		101	Mean = 7.88, SD = 1.35	4.75–11.50	—
Prenatal maternal depression		104	Median = 4, IQR = 5	0–19	84.1% below >12 cutoff (Cox et al. 1987), missing data (7.96%)
Maternal education		110	Median = "undergraduate"	IQR: "undergraduate" to "some postgraduate"	"Some/Finished high school" (2.7%), "some/finished college" (17.7%), "undergraduate" (47.8%), "some postgraduate" (29.2%), missing data (2.7%)
Household income (CAD)		112	Median = $125,000–$149,999	IQR: $75,000–$99,999 to >$175,000	<$25k (1.8%), $25–49.9k (3.5%), $50–$74.9k (4.4%), $75–$99.9k (18.6%), $100–$124.9k (18.6%), $125–$149.9k (5.3%), $150–$174.9k (15.9%), >$175,000 (31%), missing data (0.9%)

[a]The N represents the number of participants with available (observed) data; analyses were conducted on imputed datasets to account for missingness.

et al. 2016; Niño Cruz et al. 2018). However, only a few studies have examined these effects on brain activity (Labonte-Lemoyne et al. 2017) or cortical thickness (Na et al. 2022), and none to date have employed longitudinal neuroimaging methodologies nor examined the hippocampus. This underscores the novelty of our study, which investigates whether maternal physical activity during a developmentally sensitive prenatal window, the second trimester (Seress et al. 2001; White et al. 2024; Zhong et al. 2020), has lasting macrostructural or microstructural implications for hippocampal development.

We found no associations between maternal prenatal physical activity and child hippocampal structure. In contrast, Labonte-Lemoyne et al. (2017) found that maternal aerobic exercise during pregnancy was associated with more mature cortical brain activity in neonates, as measured by electroencephalography. Similarly, Na et al. (2022) found that higher physical activity in early and mid-pregnancy is linked to increased cortical thickness in newborns' brains. Supporting evidence from animal studies suggests that prenatal exercise promotes hippocampal neurogenesis and increases BDNF expression in offspring (Akhavan et al. 2008; Bick-Sander et al. 2006; Dayi et al. 2012; Gobinath et al. 2018; Gomes da Silva et al. 2016; Ji et al. 2020; Lee et al. 2006). The lack of significant effects in our study may reflect methodological or biological factors that obscure these associations in humans.

One important consideration is the type and intensity of physical activity or exercise performed during pregnancy. For example, evidence suggests that certain forms of exercise, such as resistance training, may improve cognitive functions without affecting hippocampal structure (Clark et al. 2017). Voss et al. (2019) highlight that aerobic exercise, especially at moderate to high intensity and sustained over time, is most consistently linked to hippocampal plasticity and memory benefits. In contrast, resistance and multimodal training may exert distinct or complementary effects through separate molecular mechanisms, often without structural brain changes (Voss et al. 2019). All cited rodent studies examined the effects of aerobic exercise—such as swimming, wheel running, and treadmill activity—on the hippocampus (Akhavan et al. 2008; Bick-Sander et al. 2006; Dayi et al. 2012; Gobinath et al. 2018; Gomes da Silva et al. 2016; Ji et al. 2020; Lee et al. 2006). In contrast, our study used the validated Baecke Physical Activity Questionnaire, which does not distinguish between exercise modalities. As a result, the reported activities may have been insufficient in intensity or lacked the aerobic component required to affect hippocampal development.

Previous exercise-based intervention studies have reported small to moderate improvements in brain and cognitive outcomes, such as executive network function in older adults (Voss et al. 2010 [η^2 = 0.10]) and language development in children (Leão et al. 2022). Similarly, meta-analyses indicate that resistance, aerobic, and combined training result in small positive effects on hippocampal volume in older adults (Wilckens et al. 2021), whereas parental exercise in rodents yields a moderate (16.5%) increase in offspring neurogenesis (Yang et al. 2021). These findings suggest that even structured, moderate-to-high-intensity exercise typically leads to only modest enhancements in hippocampal morphology, neural function, and behavioral outcomes. Consequently, the comparatively lower intensity and heterogeneous physical activities quantified via the Baecke Physical Activity

TABLE 2 | Longitudinal generalized fractional polynomial mixed models (GFPMM) of age-, sex-related effects on the hippocampal structural metrics (standardized β, [confidence intervals], p value).

Structural metrics	Hemisphere		Age (standardized β, [CIs], p)	Sex (standardized β, [CIs], p) [a]	Age × sex (standardized β, [CIs], p)	ICV (standardized β, [CIs], p)	Prenatal maternal physical activity (standardized β, [CIs], p)	Prenatal maternal depression (standardized β, [CIs], p)	Maternal education (standardized β, [CIs], p)	Household income (standardized β, [CIs], p)
Volume	Right		Age^{-2} (−0.198, [−0.26, −0.14], 5.05e − 10)	−0.060, [−0.20, 0.09], 0.406	NS	0.518, [0.39, 0.65], 2.57e − 14	0.011, [−0.16, 0.11], 0.863	−0.057, [−0.19, 0.10], 0.418	0.079, [−0.05, 0.23], 0.269	0.088, [−0.05, 0.23], 0.220
	Left		$Age^{-0.5}$ (−0.288, [−0.35, −0.21], 6.71e − 14)	−0.062, [−0.20, 0.09], 0.391	NS	0.556, [0.41, 0.70], 1.77e − 12	−0.079, [−0.22, 0.03], 0.212	−0.081, [−0.20, 0.06], 0.233	0.020, [−0.11, 0.16], 0.774	0.039, [−0.10, 0.17], 0.568
FA	Right	Whole sample	Age^{-2} (1.150, [0.502, 1.79], 0.00054), age^{-1} (−0.904, [−1.51, −0.29], 0.0038)	−0.455, [−0.72, −0.19], 0.0090	0.474, [0.22, 0.73], 0.00031	NA	0.036, [−0.12, 0.16], 0.616	0.017, [−0.13, 0.16], 0.815	0.130, [−0.02, 0.29], 0.095	0.015, [−0.13, 0.16], 0.847
		Males	$Age^{-0.5}$ (1.32, [0.70, 1.93], 2.81e − 5), $Age^{0.5}$ (1.27, [0.66, 1.88], 4.50e − 5)	NA	NA	NA	−0.014, [−0.24, 0.21], 0.90	−0.059, [−0.30, 0.18], 0.63	0.09, [−0.12, 0.31], 0.41	0.036, [−0.20, 0.27], 0.764
		Females	Age^{-2} (1.33, [0.65, 2.01], 1.34e − 4), Age^{-1} (−1.51, [−2.19, −0.83], 1.51e − 5)	NA	NA	NA	0.074, [−0.09, 0.24], 0.38	0.092, [−0.08, 0.26], 0.29	0.147, [−0.03, 0.33], 0.11	−0.02, [−0.20, 0.16], 0.81
	Left	Whole sample	Age^{-2} (0.350, [0.20, 0.50], 9.18e − 06)	−0.505, [−0.76, −0.25], 9.67e − 05	0.411, [0.17, 0.65], 7.47e − 04	NA	0.071, [−0.06, 0.21], 0.28	0.038, [−0.10, 0.17], 0.62	0.066, [−0.08, 0.21], 0.38	0.091, [−0.05, 0.23], 0.27

(Continues)

TABLE 2 | (Continued)

Structural metrics	**Hemisphere**	**Age (standardized β, [CIs], p)**	**Sex (standardized β, [CIs], p)** [a]	**Age × sex (standardized β, [CIs], p)**	**ICV (standardized β, [CIs], p)**	**Prenatal maternal physical activity (standardized β, [CIs], p)**	**Prenatal maternal depression (standardized β, [CIs], p)**	**Maternal education (standardized β, [CIs], p)**	**Household income (standardized β, [CIs], p)**
Males Age^{-2} (0.365, [0.18, 0.55], 1.04e − 4)	NA	NA	NA	0.080, [−0.12, 0.28], 0.43	0.091, [−0.13, 0.32], 0.42	0.160, [−0.06, 0.38], 0.16	0.110, [−0.12, 0.34], 0.34		
Females $Age^{-0.5}$ (−0.246, [−0.49, −0.01], 0.0450), $Age^{-0.5}$ (−0.356, [−0.63, −0.08], 0.0116)	NA	NA	NA	0.070, [−0.12, 0.26], 0.47	0.003, [−0.17, 0.17], 0.98	−0.014, [−0.21, 0.18], 0.89	0.088, [−0.09, 0.27], 0.34		
MD	Right	Age^{-2} (0.431, [0.35, 0.51], <2e − 16)	−0.197, [−0.35, −0.05], 0.012	NS	NA	−0.015, [−0.16, 0.13], 0.65	0.011, [−0.14, 0.17], 0.91	−0.008, [−0.16, 0.14], 0.91	0.027, [−0.13, 0.18], 0.69
	Left	Age^{-1} (0.374, [0.30, 0.45], <2e − 16)	−0.218, [−0.38, −0.06], 0.009	NS	NA	0.008, [−0.15, 0.17], 0.92	0.09, [−0.08, 0.25], 0.28	0.001, [−0.16, 0.16], 0.99	−0.05, [−0.21, 0.12], 0.64

(Continues)

TABLE 2 | (Continued)

Structural metrics	Hemisphere	Age (standardized β, [CIs], p)	Sex (standardized β, [CIs], p) [a]	Age × sex (standardized β, [CIs], p)	ICV (standardized β, [CIs], p)	Prenatal maternal physical activity (standardized β, [CIs], p)	Prenatal maternal depression (standardized β, [CIs], p)	Maternal education (standardized β, [CIs], p)
RD	Right	Age^{-1} (0.418, [0.34, 0.50], <2e − 16)	−0.173, [−0.33, −0.02], 0.028	NS	NA	−0.042, [−0.17, 0.13], 0.58	0.006, [−0.15, 0.17], 0.94	−0.022, [−0.17, 0.13], 0.78
	Left	$Age^{-0.5}$ (0.338, [0.26, 0.42], 1.91e − 15)	−0.169, [−0.34, −0.002], 0.052	NS	NA	−0.004, [−0.17, 0.16], 0.96	0.089, [−0.09, 0.26], 0.31	−0.011, [−0.17, 0.15], 0.89

Abbreviations: FA, fractional anisotropy; ICV, intracranial volume; MD, mean diffusivity; RD, radial diffusivity.

[a]Positive standardized βs represent higher values in females. In GFPMM analyses, age × sex interaction was included ($p < 0.05$) but removed from the model if not significant (denoted by NS). NA indicates parameters that were not included in the model.

TABLE 3 | Cross-sectional multivariable fractional polynomial (MFP) regression models of age-, sex-related effects on the hippocampal structural metrics (standardized β, [confidence intervals], p value).

Structural metrics	Hemisphere	Age (standardized β, [CIs], p)	Sex (standardized β, [CIs], p) [a]	Age × sex (standardized β, [CIs], p)	ICV (standardized β, [CIs], p)	Prenatal maternal physical activity (standardized β, [CIs], p)	Prenatal maternal depression (standardized β, [CIs], p)	Maternal education (standardized β, [CIs], p)	Household income (standardized β, [CIs], p)
Volume	Right	Age^1 (0.087, [−0.088, 0.262], 0.325)	0.002, [−0.207, 0.211], 0.985	NS	0.565, [0.340, 0.789], 2.95e − 6	−0.041, [−0.219, 0.138], 0.651	−0.030, [−0.216, 0.156], 0.749	0.107, [−0.075, 0.289], 0.246	0.171, [−0.015, 0.357], 0.071
	Left	Age^1 (0.232, [0.055, 0.408], 0.011)	0.155, [−0.056, 0.366], 0.149	NS	0.584, [0.356, 0.811], 1.88e − 6	−0.087, [−0.267, 0.094], 0.344	−0.005, [−0.193, 0.183], 0.958	0.043, [−0.141, 0.227], 0.640	0.083, [−0.105, 0.271], 0.385
FA	Right	Age^1 (−0.274, [−0.488, −0.060], 0.013	−0.164, [−0.369, 0.040], 0.115	NS	NA	−0.039, [−0.242, 0.164], 0.703	−0.164, [−0.373, 0.044], 0.121	0.116, [−0.099, 0.331], 0.287	−0.001, [−0.232, 0.231], 0.995
	Left	Age^1 (−0.167, [−0.378, 0.043], 0.118)	−0.257, [−0.457, −0.056], 0.013	NS	NA	0.080, [−0.119, 0.280], 0.426	−0.039, [−0.244, 0.166], 0.704	−0.016, [−0.228, 0.195], 0.879	0.272, [0.045, 0.500], 0.019
MD	Right	Age^1 (−0.319, [−0.526, −0.112], $p = 0.003$)	−0.279, [−0.476, −0.081], 0.006	NS	NA	−0.041, [−0.238, 0.156], 0.680	0.023, [−0.179, 0.225], 0.821	0.024, [−0.184, 0.232], 0.819	0.048, [−0.176, 0.271], 0.674
	Left	Age^1 (−0.225, [−0.438, −0.012], 0.038)	−0.258, [−0.462, −0.055], 0.013	NS	NA	−0.036, [−0.238, 0.167], 0.727	0.041, [−0.166, 0.249], 0.692	0.052, [−0.161, 0.266], 0.627	0.127, [−0.103, 0.357], 0.275

(Continues)

TABLE 3 | (Continued)

Structural metrics	Hemisphere	Age (standardized β, [CIs], p)	Sex (standardized β, [CIs], p) [a]	Age × sex (standardized β, [CIs], p)	ICV (standardized β, [CIs], p)	Prenatal maternal physical activity (standardized β, [CIs], p)	Prenatal maternal depression (standardized β, [CIs], p)	Maternal education (standardized β, [CIs], p)	Household income (standardized β, [CIs], p)
RD	Right	Age[1] (−0.275, [−0.489, −0.061], 0.012)	−0.250, [−0.454, −0.046], 0.017	NS	NA	−0.033, [−0.236, 0.170], 0.747	0.054, [−0.155, 0.262], 0.608	0.003, [−0.21, 0.22], 0.98	0.051, [−0.18, 0.28], 0.66
	Left	Age[1] (−0.166, [−0.390, 0.058], 0.144)	−0.179, [−0.393, 0.0344], 0.099	NS	NA	−0.050, [−0.263, 0.163], 0.642	0.051, [−0.167, 0.270], 0.642	0.049, [−0.176, 0.274], 0.664	0.054, [−0.188, 0.296], 0.657

Abbreviations: FA, fractional anisotropy; ICV, intracranial volume; MD, mean diffusivity; RD, radial diffusivity.

[a]Positive standardized βs represent higher values in females. In MFP analyses, age × sex interaction was included ($p < 0.05$) but removed from the model if not significant (denoted by NS). NA indicates parameters that were not included in the model.

Questionnaire may engender structural effects of insufficient magnitude to be detectable. Such subtle alterations may fall beneath the sensitivity threshold of volumetric analyses, particularly when restricted to global hippocampal metrics.

We also addressed the potential influence of the "physical activity paradox," whereby occupational physical activity is often unrelated or negatively associated with health outcomes, whereas leisure-time physical activity tends to be beneficial (Holtermann et al. 2018; de Vries and Bakker 2022). By reanalyzing our models using the sport subscale of the Baecke questionnaire, which primarily captures leisure-time physical activity, we confirmed that the null findings persisted. This strengthens confidence that the absence of association is not attributable to the inclusion of occupational activity in the total score. Nonetheless, self-report questionnaires provide only approximate estimates of prenatal activity and do not capture prepregnancy levels or postpartum family lifestyle factors that may influence hippocampal development across childhood.

Another potential explanation for the null findings involves the regional specificity of exercise-induced hippocampal changes. This study analyzed the hippocampus without segmentation into subfields (e.g., CA1–CA4, DG, subiculum) or along its anteroposterior axis (e.g., head, body, and tail). Given the hippocampus' anatomical and functional heterogeneity (Duvernoy et al. 2013; Insausti and Amaral 2012), this approach may have obscured localized effects of prenatal exercise. Rodent studies support this notion, showing that maternal exercise increases neuronal cell counts in the CA1 and DG, but not the subiculum (Akhavan et al. 2008), and enhances neurogenesis in the dorsal hippocampus of adult offspring (Gobinath et al. 2018)—a region that functionally corresponds to the posterior hippocampus in humans (Fanselow and Dong 2010). Erickson et al. (2011) also found region-specific effects in humans, observing increases in anterior hippocampal volume following aerobic training in older adults. Together, these findings highlight the importance of region-specific analyses, whose absence in this study may have contributed to the lack of observed effects. Lastly, although maternal physical activity data were available for the first trimester in a smaller subsample ($n = 53$), they were excluded from the primary analysis, partially due to insufficient statistical power. Visual inspection of the scatter plot showed nearly all participants exhibiting comparable activity levels across the first and second trimesters. The only notable exception was one participant whose total activity score declined from 9.625 in the first trimester to 5.875 in the second. Given this stability, it is reasonable to infer that similar null results would likely have emerged for the first trimester, had the sample size been sufficient—reinforcing the conclusion that prenatal maternal physical activity, as measured in this study, is not associated with hippocampal volume in children.

Maternal depressive symptoms—whether clinical or subclinical—during the perinatal period and extending into early childhood are associated with adverse child outcomes, including alterations in amygdala microstructure, cortical thickness, and stress-related systems, as well as increased risk of emotional–behavioral difficulties (Giallo et al. 2015; Meaney 2018). Similarly, Lebel et al. (2016) demonstrated that both prenatal and postpartum maternal depressive symptoms are associated with altered cortical and white matter development in preschool-aged children, suggesting potentially accelerated maturation in brain regions critical for executive function and emotional regulation. Previous neuroimaging studies examining the effects of prenatal maternal depression on the hippocampus have primarily focused on fetuses (Wu et al. 2020), neonates and infants (Acosta et al. 2020; Groenewold et al. 2022; Qiu et al. 2017), or postnatal maternal depression and hippocampal structure in older children and adolescents (Chen et al. 2010; Gilliam et al. 2015; Hubachek et al. 2021; Lupien et al. 2011). An exception is a prior longitudinal study from our group, based on a largely overlapping dataset (Donnici et al. 2023), which similarly found no significant associations between prenatal maternal depressive symptoms and hippocampal volume. Evidence from Qiu et al. (2017), Acosta et al. (2020), and Groenewold et al. (2022) indicates that prenatal maternal depressive symptoms influence hippocampal structure in neonates and infants, with effects moderated by genetic predisposition and infant sex. Qiu et al. (2017) reported increased right hippocampal volume and altered anterior hippocampal shape in neonates with elevated genetic risk for depression; Acosta et al. (2020) found reduced right hippocampal volume in high-risk male infants; and Groenewold et al. (2022) observed enlarged hippocampal volumes in female infants exposed to maternal depressive symptoms. Wu et al. (2020) extended this evidence to the fetal period, showing that prenatal maternal psychological distress—especially anxiety and depression—was linked to reduced fetal hippocampal volume, especially in the left hemisphere. Taken together, the current evidence suggests that prenatal maternal depression exerts its most discernible effects on hippocampal structure before and shortly after birth, with the direction and magnitude of change moderated by fetal sex and genetic risk. By early childhood, these volumetric differences are no longer evident, underscoring the plasticity of hippocampal development. Consistent with this interpretation, animal models of depression have demonstrated that stress-induced dendritic changes in the hippocampal CA3 region can be fully reversed following post-stress enrichment, such as water-maze training (Qiao et al. 2016). This finding aligns with the absence of hippocampal volume differences observed from early childhood through adolescence in our longitudinal human cohort. Another potential explanation for our null findings is the use of an undifferentiated hippocampal region of interest. Neuroimaging studies in major depressive disorder have demonstrated that volumetric reductions are confined to discrete subfields—namely, CA1–3 and the DG—primarily in the posterior hippocampus (Malykhin and Coupland 2015). Hubachek et al. (2021) further reported subregion-specific effects in preadolescents of depressed mothers. Similarly, rodent models of chronic stress and depression reveal subfield-specific dendritic retraction, particularly in the dorsal CA3 (Conrad et al. 2017; Qiao et al. 2016). Aggregating all subfields and subregions into a single volume metric may obscure such localized effects. Moreover, only 7.96% of mothers in our cohort met or exceeded the EPDS ≥12 threshold, indicating a low prevalence of clinically significant symptoms, which may have further attenuated observable effects on hippocampal structure.

We found no significant associations between maternal education or household income and hippocampal structural metrics. However, maternal education showed a weak positive association with right hippocampal volume in the overall sample (p: 0.095), although the evidence was inconclusive. Emerging evidence suggests that maternal and parental education play a key role in hippocampal development. Alex et al. (2024) reported that lower maternal education was significantly linked to smaller hippocampal volume in early childhood. Similarly, Noble et al. (2015) found a significant, nonlinear association between parental education and left hippocampal volume, with increases in hippocampal volume per year of parental education being greatest among children whose parents had lower levels of education. Extending these findings into adolescence, Ellwood-Lowe et al. (2018) showed that higher parental education was consistently associated with larger hippocampal volume throughout adolescence, and that this relationship remained significant even after accounting for maternal brain structure. This indicates that environmental influences related to parental education play a distinct role in shaping hippocampal development.

A plausible explanation for the null findings in our study is that the predominance of highly educated mothers in our analyzed cohort (approximately 77% with an undergraduate degree or higher), which may attenuate any true association with offspring hippocampal volume via a ceiling effect (Garin 2014). This limited variability reduces statistical power to identify maternal education-related influences on hippocampal development, even if they are present.

Previous studies of hippocampal development in childhood and adolescence have produced varied findings, possibly reflecting methodological and analytical differences. Early cross-sectional research using automated segmentation techniques reported nonlinear age-related increases in ICV-adjusted hippocampal volume during childhood and adolescence (Østby et al. 2009). However, a subsequent longitudinal study of this cohort did not observe volumetric changes at the individual level (Tamnes et al. 2013). Other longitudinal studies describe an inverted-U trajectory for hippocampal growth (not adjusted for ICV), further complicating interpretation (Wierenga et al. 2014; Goddings et al. 2014). More precise manual segmentation techniques provided evidence of subfield-specific changes; Lee et al. (2014) reported cross-sectional age-related differences, with larger CA3/DG and CA1 subfield volumes in older children and adolescents and no significant change in the subiculum. Gogtay et al. (2006) observed regionally varying changes over time, with volumetric increases in the hippocampal body and decreases in the head and tail. Similarly, Canada et al. (2020) found that the best fitting model for bilateral total hippocampal volume was cubic, with a quadratic growth observed in the head and linear increases in the body and tail among younger children. Our study revealed nonlinear increases in both right and left hippocampal volumes, with age-related plateaus observed in the right hippocampus. Additionally, ICV emerged as a robust predictor in our study, reinforcing the importance of adjusting hippocampal volumes for ICV in analyses. Collectively, these findings underscore the complexity of hippocampal development, emphasizing that future longitudinal investigations integrating detailed cytoarchitectural and anterior–posterior subdivisions are crucial for clarifying how hippocampal structural maturation relates to cognitive development across childhood and adolescence (Lee et al. 2017).

There are few limitations that warrant consideration and should be addressed in future studies. First, our use of a global hippocampal volume may have obscured potential subfield- and subregion-specific effects, as evidence from human and animal models emphasizes focal vulnerability within CA1–3, DG, and posterior segments. Second, the high socioeconomic status of our cohort may have introduced ceiling effects and limited variability, reducing the power to detect associations. Finally, prenatal physical activity was measured using the self-reported Baecke Physical Activity Questionnaire, which provides broad domain scores but does not precisely capture activity intensity, duration, or exercise modality (e.g., aerobic vs. resistance). As a result, the "physical activity" quantified in this study may have been too heterogeneous or insufficiently intense to produce detectable hippocampal structural differences. Although we re-analyzed models using the sport subscale (to better reflect leisure-time activity), results remained null, suggesting the absence of association is unlikely to be driven solely by inclusion of occupational activity in the total score.

5 | Conclusion

Leveraging a large MRI dataset and rigorous GFPMM modeling, this accelerated longitudinal study detailed hippocampal macrostructural and microstructural trajectories from early childhood through adolescence. We identified nonlinear volumetric growth and maturation of diffusion metrics, with sex differences confined to FA. Despite the statistical power afforded by our comprehensive dataset and advanced modeling, no associations emerged between total hippocampal volume, anisotropy, or diffusivity and prenatal maternal physical activity, depressive symptoms, or education. In our generally healthy, high-socioeconomic-status cohort, prenatal exposures quantified by the Baecke Physical Activity Questionnaire and Edinburgh Postnatal Depression Scale exerted negligible influence on global hippocampal structure beyond normative developmental trajectories.

Author Contributions

Arash Aghamohammadi-Sereshki and Merv Singh conducted the data analyses and drafted the manuscript. Jess E. Reynolds contributed to the original study design, data curation and processing, and funding acquisition. Jamie Roeske contributed to data curation and processing. Rhonda C. Bell, Laura Forbes, Gerald F. Giesbrecht, and Nicole Letourneau contributed to the original APrON study design. Deborah Dewey contributed to the study design of both the APrON study and the present study. Catherine Lebel contributed to the study design, supervised the present study, and secured funding. All authors critically reviewed and edited the manuscript and approved the final version for publication.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Open Science Framework at https://osf.io/axz5r/.

References

Acosta, H., K. Kantojärvi, N. Hashempour, et al. 2020. "Partial Support for an Interaction Between a Polygenic Risk Score for Major Depressive Disorder and Prenatal Maternal Depressive Symptoms on Infant Right Amygdalar Volumes." *Cerebral Cortex* 30, no. 12: 6121–6134. https://doi.org/10.1093/cercor/bhaa158.

Aghamohammadi-Sereshki, A., F. Olsen, P. Seres, and N. V. Malykhin. 2022. "Selective Effects of Healthy Cognitive Aging and Catechol-O-Methyl Transferase Polymorphism on Limbic White Matter Tracts." *Brain Connect* 12, no. 2: 146–163. https://doi.org/10.1089/brain.2020.0919.

Akhavan, M. M., M. Emami-Abarghoie, M. Safari, et al. 2008. "Serotonergic and Noradrenergic Lesions Suppress the Enhancing Effect of Maternal Exercise During Pregnancy on Learning and Memory in Rat Pups." *Neuroscience* 151, no. 4: 1173–1183. https://doi.org/10.1016/j.neuroscience.2007.10.051.

Alex, A. M., F. Aguate, K. Botteron, et al. 2024. "A Global Multicohort Study to Map Subcortical Brain Development and Cognition in Infancy and Early Childhood." *Nature Neuroscience* 27: 176–186. https://doi.org/10.1038/s41593-023-01501-6.

Ambler, G., and A. Benner. 2024. *Mfp: Multivariable Fractional Polynomials (Version 1.5.4.1) [R Package]*. CRAN. https://doi.org/10.32614/CRAN.package.mfp.

Andersson, J. L. R., M. S. Graham, E. Zsoldos, and S. N. Sotiropoulos. 2016. "Incorporating Outlier Detection and Replacement Into a Non-Parametric Framework for Movement and Distortion Correction of Diffusion MR Images." *Neuroimage* 141: 556–572. https://doi.org/10.1016/j.neuroimage.2016.06.058.

Andersson, J. L. R., and S. N. Sotiropoulos. 2016. "An Integrated Approach to Correction for Off-Resonance Effects and Subject Movement in Diffusion MR Imaging." *Neuroimage* 125: 1063–1078. https://doi.org/10.1016/j.neuroimage.2015.10.019.

Alves, J. M., S. Luo, T. Chow, M. Herting, A. H. Xiang, and K. A. Page. 2020. "Sex Differences in the Association Between Prenatal Exposure to Maternal Obesity and Hippocampal Volume in Children." *Brain and Behavior* 10, no. 2: e01522. https://doi.org/10.1002/brb3.1522.

Asman, A. J., and B. A. Landman. 2012. "Formulating Spatially Varying Performance in the Statistical Fusion Framework." *IEEE Transactions on Medical Imaging* 31, no. 6: 1326–1336. https://doi.org/10.1109/TMI.2012.2190992.

Asman, A. J., and B. A. Landman. 2013. "Non-Local Statistical Label Fusion for Multi-Atlas Segmentation." *Medical Image Analysis* 17, no. 2: 194–208. https://doi.org/10.1016/j.media.2012.10.002.

Austin, P. C., I. R. White, D. S. Lee, and S. van Buuren. 2021. "Missing Data in Clinical Research: A Tutorial on Multiple Imputation." *Canadian Journal of Cardiology* 37, no. 9: 1322–1331. https://doi.org/10.1016/j.cjca.2020.11.010.

Baecke, J. A., J. Burema, and J. E. Frijters. 1982. "A Short Questionnaire for the Measurement of Habitual Physical Activity in Epidemiological Studies." *American Journal of Clinical Nutrition* 36, no. 5: 936–942. https://doi.org/10.1093/ajcn/36.5.936.

Bastiani, M., M. Cottaar, S. P. Fitzgibbon, et al. 2019. "Automated Quality Control for Within and Between Studies Diffusion MRI Data Using a Non-Parametric Framework for Movement and Distortion Correction." *Neuroimage* 184: 801–812. https://doi.org/10.1016/j.neuroimage.2018.09.073.

Bates, D., M. Mächler, B. Bolker, and S. Walker. 2015. "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software* 67, no. 1: 1–48. https://doi.org/10.18637/jss.v067.i01.

Bazin, P.-L., and D. L. Pham. 2008. "Homeomorphic Brain Image Segmentation With Topological and Statistical Atlases." *Medical Image Analysis* 12, no. 5: 616–625. https://doi.org/10.1016/j.media.2008.06.008.

Bergink, V., L. Kooistra, M. P. Lambregtse-van den Berg, et al. 2011. "Validation of the Edinburgh Depression Scale During Pregnancy." *Journal of Psychosomatic Research* 70, no. 4: 385–389. https://doi.org/10.1016/j.jpsychores.2010.07.008.

Bermudez, C., J. Blaber, S. W. Remedios, et al. 2020. "Generalizing Deep Whole Brain Segmentation for Pediatric and Post- Contrast MRI With Augmented Transfer Learning." *Proceedings of SPIE* 11313: 113130L. https://pubmed.ncbi.nlm.nih.gov/34040280/.

Bick-Sander, A., B. Steiner, S. A. Wolf, H. Babu, and G. Kempermann. 2006. "Running in Pregnancy Transiently Increases Postnatal Hippocampal Neurogenesis in the Offspring." *Proceedings of the National Academy of Sciences* 103, no. 10: 3852–3857. https://doi.org/10.1073/pnas.0502644103.

Burdette, J. H., P. J. Laurienti, M. A. Espeland, et al. 2010. "Using Network Science to Evaluate Exercise-Associated Brain Changes in Older Adults." *Frontiers in Aging Neuroscience* 2: 23. https://doi.org/10.3389/fnagi.2010.00023.

Canada, K. L., M. Botdorf, and T. Riggins. 2020. "Longitudinal Development of Hippocampal Subregions From Early- to Mid-Childhood." *Hippocampus* 30, no. 10: 1098–1111. https://doi.org/10.1002/hipo.23218.

Center on the Developing Child at Harvard University. 2009. "Maternal Depression Can Undermine the Development of Young Children." Working Paper No. 8. Center on the Developing Child at Harvard University. http://www.developingchild.harvard.edu.

Chen, M. C., J. P. Hamilton, and I. H. Gotlib. 2010. "Decreased Hippocampal Volume in Healthy Girls at Risk of Depression." *Archives of General Psychiatry* 67, no. 3: 270–276. https://doi.org/10.1001/archgenpsychiatry.2009.202.

Clark, R., C. Wendel, and M. W. Voss. 2017. "Physical Activity and Cognitive Training: Impact on Hippocampal Structure and Function." In *The Hippocampus From Cells to Systems*, edited by D. E. Hannula and M. C. Duff, 209–243. Springer. https://doi.org/10.1007/978-3-319-50406-3_8.

Conrad, C. D., J. B. Ortiz, and J. M. Judd. 2017. "Chronic Stress and Hippocampal Dendritic Complexity: Methodological and Functional Considerations." *Physiology & Behavior* 178: 66–81. https://doi.org/10.1016/j.physbeh.2016.11.017.

Cox, J. L., J. M. Holden, and R. Sagovsky. 1987. "Detection of Postnatal Depression. Development of the 10-Item Edinburgh Postnatal Depression Scale." *British Journal of Psychiatry* 150: 782–786. https://doi.org/10.1192/bjp.150.6.782.

Cox, R. W. 1996. "AFNI: Software for Analysis and Visualization of Functional Magnetic Resonance Neuroimages." *Computers and Biomedical Research* 29, no. 3: 162–173. https://doi.org/10.1006/cbmr.1996.0014.

Cox, R. W., and J. S. Hyde. 1997. "Software Tools for Analysis and Visualization of fMRI Data." *NMR in Biomedicine* 10, no. 4–5: 171–178. https://doi.org/10.1002/(sici)1099-1492(199706/08)10:4/5⟨171::aid-nbm453⟩3.0.co;2-l.

Dayi, A., S. Agilkaya, S. Ozbal, et al. 2012. "Maternal Aerobic Exercise During Pregnancy Can Increase Spatial Learning by Affecting Leptin Expression on Offspring's Early and Late Period in Life Depending on Gender." *Scientific World Journal* 2012: 429803. https://doi.org/10.1100/2012/429803.

de Vries, J. D., and A. B. Bakker. 2022. "The Physical Activity Paradox: A Longitudinal Study of the Implications for Burnout." *International Archives of Occupational and Environmental Health* 95, no. 5: 965–979. https://doi.org/10.1007/s00420-021-01759-y.

Donnici, C., X. Long, J. Reynolds, et al. 2023. "Prenatal Depressive Symptoms and Childhood Development of Brain Limbic and Default Mode Network Structure." *Human Brain Mapping* 44, no. 6: 2380–2394. https://doi.org/10.1002/hbm.26216.

Duvernoy, H. M., F. Cattin, and P.-Y. Risold. 2013. *The Human Hippocampus: Functional Anatomy, Vascularization and Serial Sections With MRI*. 4th ed. Springer. https://doi.org/10.1007/978-3-642-33603-4.

Ellwood-Lowe, M. E., K. L. Humphreys, S. J. Ordaz, M. C. Camacho, M. D. Sacchet, and I. H. Gotlib. 2018. "Time-Varying Effects of Income on Hippocampal Volume Trajectories in Adolescent Girls." *Developmental Cognitive Neuroscience* 30: 41–50. https://doi.org/10.1016/j.dcn.2017.12.005.

Erickson, K. I., M. W. Voss, R. S. Prakash, et al. 2011. "Exercise Training Increases Size of Hippocampus and Improves Memory." *Proceedings of the National Academy of Sciences* 108, no. 7: 3017–3022. https://doi.org/10.1073/pnas.1015950108.

Fanselow, M. S., and H. W. Dong. 2010. "Are the Dorsal and Ventral Hippocampus Functionally Distinct Structures?" *Neuron* 65, no. 1: 7–19. https://doi.org/10.1016/j.neuron.2009.11.031.

Fjell, A. M., K. B. Walhovd, L. T. Westlye, et al. 2010. "When Does Brain Aging Accelerate? Dangers of Quadratic Fits in Cross-Sectional Studies." *Neuroimage* 50, no. 4: 1376–1383. https://doi.org/10.1016/j.neuroimage.2010.01.061.

Garin, O. 2014. "Ceiling Effect." In *Encyclopedia of Quality of Life and Well-Being Research*, edited by A. C. Michalos, 796–798. Springer. https://doi.org/10.1007/978-94-007-0753-5_296.

Giallo, R., H. Woolhouse, D. Gartland, H. Hiscock, and S. Brown. 2015. "The Emotional-Behavioural Functioning of Children Exposed to Maternal Depressive Symptoms Across Pregnancy and Early Childhood: A Prospective Australian Pregnancy Cohort Study." *European Child & Adolescent Psychiatry* 24, no. 10: 1233–1244. https://doi.org/10.1007/s00787-014-0672-2.

Gilliam, M., E. E. Forbes, P. J. Gianaros, K. I. Erickson, L. M. Brennan, and D. S. Shaw. 2015. "Maternal Depression in Childhood and Aggression in Young Adulthood: Evidence for Mediation by Offspring Amygdala-Hippocampal Volume Ratio." *Journal of Child Psychology and Psychiatry* 56, no. 10: 1083–1091. https://doi.org/10.1111/jcpp.12364.

Gobinath, A. R., S. Wong, C. Chow, S. E. Lieblich, A. M. Barr, and L. A. M. Galea. 2018. "Maternal Exercise Increases but Concurrent Maternal Fluoxetine Prevents the Increase in Hippocampal Neurogenesis of Adult Offspring." *Psychoneuroendocrinology* 91: 186–197. https://doi.org/10.1016/j.psyneuen.2018.02.027.

Goddings, A. L., K. L. Mills, L. S. Clasen, J. N. Giedd, R. M. Viner, and S. J. Blakemore. 2014. "The Influence of Puberty on Subcortical Brain Development." *Neuroimage* 88: 242–251. https://doi.org/10.1016/j.neuroimage.2013.09.073.

Gogtay, N., T. F. Nugent 3rd, D. H. Herman, et al. 2006. "Dynamic Mapping of Normal Human Hippocampal Development." *Hippocampus* 16, no. 8: 664–672. https://doi.org/10.1002/hipo.20193.

Gomes da Silva, S., A. A. de Almeida, J. Fernandes, et al. 2016. "Maternal Exercise During Pregnancy Increases BDNF Levels and Cell Numbers in the Hippocampal Formation but Not in the Cerebral Cortex of Adult Rat Offspring." *PLoS ONE* 11, no. 1: e0147200. https://doi.org/10.1371/journal.pone.0147200.

Groenewold, N. A., C. J. Wedderburn, J. A. Pellowski, et al. 2022. "Subcortical Brain Volumes in Young Infants Exposed to Antenatal Maternal Depression: Findings From a South African Birth Cohort." *NeuroImage: Clinical* 36: 103206. https://doi.org/10.1016/j.nicl.2022.103206.

Herba, C. M., V. Glover, P. G. Ramchandani, and M. B. Rondon. 2016. "Maternal Depression and Mental Health in Early Childhood: An Examination of Underlying Mechanisms in Low-Income and Middle-Income Countries." *Lancet Psychiatry* 3, no. 10: 983–992. https://doi.org/10.1016/S2215-0366(16)30148-1.

Holtermann, A., N. Krause, A. J. van der Beek, and L. Straker. 2018. "The Physical Activity Paradox: Six Reasons Why Occupational Physical Activity (OPA) Does Not Confer the Cardiovascular Health Benefits That Leisure Time Physical Activity Does." *British Journal of Sports Medicine* 52: 149–150. https://doi.org/10.1136/bjsports-2017-097965.

Hubachek, S., M. Botdorf, T. Riggins, H. C. Leong, D. N. Klein, and L. R. Dougherty. 2021. "Hippocampal Subregion Volume in High-Risk Offspring Is Associated With Increases in Depressive Symptoms Across the Transition to Adolescence." *Journal of Affective Disorders* 281: 358–366. https://doi.org/10.1016/j.jad.2020.12.017.

Huo, Y., A. Carass, S. M. Resnick, D. L. Pham, J. L. Prince, and B. A. Landman. 2016. "Combining Multi-Atlas Segmentation With Brain Surface Estimation." *Proceedings of SPIE* 9784: 97840E. https://doi.org/10.1117/12.2216604.

Huo, Y., A. J. Plassard, A. Carass, et al. 2016. "Consistent Cortical Reconstruction and Multi-Atlas Brain Segmentation." *Neuroimage* 138: 197–210. https://doi.org/10.1016/j.neuroimage.2016.05.030.

Huo, Y., S. Bao, P. Parvathaneni, and B. A. Landman. 2018. "Improved Stability of Whole Brain Surface Parcellation With Multi-Atlas Segmentation." *Proceedings of SPIE* 10574: 1057438. https://doi.org/10.1117/12.2281509.

Insausti, R., and D. G. Amaral. 2012. "Hippocampal Formation." In *The Human Nervous System*, 3rd ed., edited by J. Mai and G. Paxinos, 897–933. Academic Press. https://doi.org/10.1016/B978-0-12-374236-0.10024-0.

Jakobsen, J. C., C. Gluud, J. Wetterslev, and P. Winkel. 2017. "When and How Should Multiple Imputation be Used for Handling Missing Data in Randomised Clinical Trials—A Practical Guide With Flowcharts." *BMC Medical Research Methodology* [Electronic Resource] 17, no. 1: 162. https://doi.org/10.1186/s12874-017-0442-1.

Ji, E. S., Y. M. Kim, Y. J. Ko, and S. S. Baek. 2020. "Treadmill Exercise in Obese Maternal Rats During Pregnancy Improves Short-Term Memory Through Neurogenesis in the Hippocampus of Rat Pups." *Journal of Exercise Rehabilitation* 16, no. 5: 392–397. https://doi.org/10.12965/jer.2040618.309.

Kellner, E., B. Dhital, V. G. Kiselev, and M. Reisert. 2016. "Gibbs-Ringing Artifact Removal Based on Local Subvoxel-Shifts." *Magnetic Resonance in Medicine* 76, no. 5: 1574–1581. https://doi.org/10.1002/mrm.26054.

Klein, A., T. Dal Canton, S. S. Ghosh, B. Landman, J. Lee, and A. Worth. 2010. "Open Labels: Online Feedback for a Public Resource of Manually Labeled Brain Images." In *Proceedings of the 16th Annual Meeting for the Organization of Human Brain Mapping*.

Kuznetsova, A., P. B. Brockhoff, and R. H. B. Christensen. 2017. "lmerTest Package: Tests in Linear Mixed Effects Models." *Journal of Statistical Software* 82, no. 13: 1–26. https://doi.org/10.18637/jss.v082.i13.

Labonte-Lemoyne, E., D. Curnier, and D. Ellemberg. 2017. "Exercise During Pregnancy Enhances Cerebral Maturation in the Newborn: A Randomized Controlled Trial." *Journal of Clinical and Experimental Neuropsychology* 39, no. 4: 347–354. https://doi.org/10.1080/13803395.2016.1227427.

Leão, O. A. A., M. R. Domingues, A. D. Bertoldi, et al. 2022. "Effects of Regular Exercise During Pregnancy on Early Childhood Neurodevelopment: The Physical Activity for Mothers Enrolled in Longitudinal Analysis Randomized Controlled Trial." *Journal of Physical Activity and Health* 19, no. 3: 203–210. https://doi.org/10.1123/jpah.2021-0477.

Lebel, C., S. Treit, and C. Beaulieu. 2019. "A Review of Diffusion MRI of Typical White Matter Development From Early Childhood to Young Adulthood." *NMR in Biomedicine* 32, no. 4: e3778. https://doi.org/10.1002/nbm.3778.

Lebel, C., M. Walton, N. Letourneau, G. F. Giesbrecht, B. J. Kaplan, and D. Dewey. 2016. "Prepartum and Postpartum Maternal Depressive Symptoms Are Related to Children's Brain Structure in Preschool." *Biological Psychiatry* 80, no. 11: 859–868. https://doi.org/10.1016/j.biopsych.2015.12.004.

Lee, H. H., H. Kim, J. W. Lee, et al. 2006. "Maternal Swimming During Pregnancy Enhances Short-Term Memory and Neurogenesis in the Hippocampus of Rat Pups." *Brain and Development* 28, no. 3: 147–154. https://doi.org/10.1016/j.braindev.2005.05.007.

Lee, J. K., A. D. Ekstrom, and S. Ghetti. 2014. "Volume of Hippocampal Subfields and Episodic Memory in Childhood and Adolescence." *Neuroimage* 94: 162–171. https://doi.org/10.1016/j.neuroimage.2014.03.019.

Lee, J. K., E. G. Johnson, and S. Ghetti. 2017. "Hippocampal Development: Structure, Function and Implications." In *The Hippocampus From Cells to Systems: Structure, Connectivity, and Functional Contributions to Memory and Flexible Cognition*, edited by D. E. Hannula and M. C. Duff, 141–166. Springer. https://doi.org/10.1007/978-3-319-50406-3_6.

Letourneau, N., F. Aghajafari, R. C. Bell, et al. 2022. "The Alberta Pregnancy Outcomes and Nutrition (APrON) Longitudinal Study: Cohort Profile and Key Findings From the First Three Years." *BMJ Open* 12, no. 2: e047503. https://doi.org/10.1136/bmjopen-2020-047503.

Long, M., C. Ostertag, J. E. Reynolds, et al. 2024. "Few Sex Differences in Regional Gray Matter Volume Growth Trajectories Across Early Childhood." *Imaging Neuroscience* 2: 1–26. https://doi.org/10.1162/imag_a_00154.

Lupien, S. J., R. P. Juster, C. Raymond, and M. F. Marin. 2018. "The Effects of Chronic Stress on the Human Brain: From Neurotoxicity, to Vulnerability, to Opportunity." *Frontiers in Neuroendocrinology* 49: 91–105. https://doi.org/10.1016/j.yfrne.2018.02.001.

Lupien, S. J., S. Parent, A. C. Evans, et al. 2011. "Larger Amygdala but No Change in Hippocampal Volume in 10-Year-Old Children Exposed to Maternal Depressive Symptomatology Since Birth." *Proceedings of the National Academy of Sciences* 108, no. 34: 14324–14329. https://doi.org/10.1073/pnas.1105371108.

Malykhin, N. V., and N. J. Coupland. 2015. "Hippocampal Neuroplasticity in Major Depressive Disorder." *Neuroscience* 309: 200–213. https://doi.org/10.1016/j.neuroscience.2015.04.047.

Marques, A. H., A. L. Bjørke-Monsen, A. L. Teixeira, and M. N. Silverman. 2015. "Maternal Stress, Nutrition and Physical Activity: Impact on Immune Function, CNS Development and Psychopathology." *Brain Research* 1617: 28–46. https://doi.org/10.1016/j.brainres.2014.10.051.

Meaney, M. J. 2018. "Perinatal Maternal Depressive Symptoms as an Issue for Population Health." *American Journal of Psychiatry* 175, no. 11: 1084–1093. https://doi.org/10.1176/appi.ajp.2018.17091031.

Moyer, C., O. R. Reoyo, and L. May. 2016. "The Influence of Prenatal Exercise on Offspring Health: A Review." *Clinical Medicine Insights: Women's Health* 9: 37–42. https://doi.org/10.4137/CMWH.S34670.

Mundfrom, D., J. Perrett, J. Schaffer, A. Piccone, and M. Roozeboom. 2006. "Bonferroni Adjustments in Tests for Regression Coefficients." *General Linear Model Journal* 32: 1–6.

Na, X., R. Raja, N. E. Phelan, et al. 2022. "Mother's Physical Activity During Pregnancy and Newborn's Brain Cortical Development." *Frontiers in Human Neuroscience* 16: 943341. https://doi.org/10.3389/fnhum.2022.943341.

Niño Cruz, G. I., A. Ramirez Varela, I. C. M. da Silva, P. C. Hallal, and I. S. Santos. 2018. "Physical Activity During Pregnancy and Offspring Neurodevelopment: A Systematic Review." *Paediatric and Perinatal Epidemiology* 32, no. 4: 369–379. https://doi.org/10.1111/ppe.12472.

Noble, K., S. Houston, N. Brito, et al. 2015. "Family Income, Parental Education and Brain Structure in Children and Adolescents." *Nature Neuroscience* 18: 773–778. https://doi.org/10.1038/nn.3983.

Ourselin, S., A. Roche, G. Subsol, X. Pennec, and N. Ayache. 2001. "Reconstructing a 3D structure from serial histological sections." *Image and Vision Computing*, 19, no. 1–2: 25–31. https://doi.org/10.1016/S0262-8856(00)00052-4.

Ourselin, S., R. Stefanescu, and X. Pennec. 2002. Robust Registration of Multi-modal Images: Towards Real-Time Clinical Applications. In: T. Dohi, R. Kikinis (eds) *Medical Image Computing and Computer-Assisted Intervention — MICCAI 2002*. MICCAI 2002. Lecture Notes in Computer Science, 2489. Springer, Berlin, Heidelberg. https://doi.org/10.1007/3-540-45787-9_18.

Posit Team. 2025. *RStudio: Integrated Development Environment for R*. Posit Software, PBC. https://www.posit.co/.

Qiao, H., M. X. Li, C. Xu, H. B. Chen, S. C. An, and X. M. Ma. 2016. "Dendritic Spines in Depression: What We Learned From Animal Models." *Neural Plasticity* 2016: 8056370. https://doi.org/10.1155/2016/8056370.

Qiu, A., M. Shen, C. Buss, et al. 2017. "Effects of Antenatal Maternal Depressive Symptoms and Socio-Economic Status on Neonatal Brain Development Are Modulated by Genetic Risk." *Cerebral Cortex* 27, no. 5: 3080–3092. https://doi.org/10.1093/cercor/bhx065.

Østby, Y., C. K. Tamnes, A. M. Fjell, L. T. Westlye, P. Due-Tønnessen, and K. B. Walhovd. 2009. "Heterogeneity in Subcortical Brain Development: A Structural Magnetic Resonance Imaging Study of Brain Maturation From 8 to 30 Years." *Journal of Neuroscience* 29, no. 38: 11772–11782. https://doi.org/10.1523/JNEUROSCI.1242-09.2009.

Reynolds, J. E., X. Long, D. Paniukov, M. Bagshawe, and C. Lebel. 2020. "Calgary Preschool Magnetic Resonance Imaging (MRI) Dataset." *Data in Brief* 29: 105224. https://doi.org/10.1016/j.dib.2020.105224.

Reynolds, J., R. C. Bell, D. M. Dewey, C. Lebel, and A. Aghamohammadi-Sereshki. 2022. *Prenatal Maternal Physical Activity and Hippocampal Structure in Young Children*. OSF. https://doi.org/10.17605/OSF.IO/Z82PU.

Roeske, J., X. Long, M. V. Perdue, et al. 2025. "Sex Differences in Maturational Timing of Amygdala and Prefrontal Cortex Volumes and White Matter Tract Microstructure." *Developmental Cognitive Neuroscience* 74: 101568. https://doi.org/10.1016/j.dcn.2025.101568.

Rosberg, A., H. Merisaari, J. D. Lewis, et al. 2025. "Associations Between Maternal Pre-Pregnancy BMI and Mean Diffusivity of the Hippocampus and Amygdala in Infants." *International Journal of Obesity* 49: 938–941. https://doi.org/10.1038/s41366-025-01730-8.

Royston, P., and W. Sauerbrei. 2008. "MFP: Multivariable Model-Building With Fractional Polynomials." In *Multivariable Model-Building: A Pragmatic Approach to Regression Analysis Based on Fractional Polynomials for Modelling Continuous Variables*, edited by P. Royston and W. Sauerbrei, 71–98. Wiley.

Rubin, D. B. 1987, *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons, Inc.

Ryoo, J. H., J. D. Long, G. W. Welch, A. Reynolds, and S. M. Swearer. 2017. "Fitting the Fractional Polynomial Model to non-Gaussian Longitudinal Data." *Frontiers in Psychology* 8: 1431. https://doi.org/10.3389/fpsyg.2017.01431.

Sabaghi, A., A. Heirani, A. Kiani, et al. 2019. "Effects of Aerobic Exercise During Pregnancy on Neurobehavioral Performances and Serum Levels of GDNF in Adult Male Mice Offspring." *Neurophysiology* 51: 430–437. https://doi.org/10.1007/s11062-020-09839-4.

Seress, L., H. Abrahám, T. Tornóczky, and G. Kosztolányi. 2001. "Cell Formation in the Human Hippocampal Formation From Mid-Gestation to the Late Postnatal Period." *Neuroscience* 105, no. 4: 831–843. https://doi.org/10.1016/s0306-4522(01)00156-7.

Tamnes, C. K., K. B. Walhovd, H. Grydeland, et al. 2013. "Longitudinal Working Memory Development Is Related to Structural Maturation of Frontal and Parietal Cortices." *Journal of Cognitive Neuroscience* 25, no. 10: 1611–1623. https://doi.org/10.1162/jocn_a_00434.

Teicher, M. H., S. L. Andersen, A. Polcari, C. M. Anderson, C. P. Navalta, and D. M. Kim. 2003. "The Neurobiological Consequences of Early Stress and Childhood Maltreatment." *Neuroscience and Biobehavioral Reviews* 27, no. 1–2: 33–44. https://doi.org/10.1016/s0149-7634(03)00007-1.

Tournier, J.-D., R. Smith, D. Raffelt, et al. 2019. "MRtrix3: A Fast, Flexible and Open Software Framework for Medical Image Processing and Visualisation." *Neuroimage* 202: 116137. https://doi.org/10.1016/j.neuroimage.2019.116137.

Valkenborghs, S. R., P. C. Dent, and C. M. Stillman. 2022. "The Intergenerational Effects of Parental Physical Activity on Offspring Brain and Neurocognition in Humans: A Scoping Review." *Neuroscience & Biobehavioral Reviews* 143: 104953. https://doi.org/10.1016/j.neubiorev.2022.104953.

Valkenborghs, S. R., C. H. Hillman, O. Al-Iedani, et al. 2022. "Effect of High-Intensity Interval Training on Hippocampal Metabolism in Older Adolescents." *Psychophysiology* 59, no. 11: e14090. https://doi.org/10.1111/psyp.14090.

van Buuren, S., and K. Groothuis-Oudshoorn. 2011. "Mice: Multivariate Imputation by Chained Equations in R." *Journal of Statistical Software* 45, no. 3: 1–67. https://doi.org/10.18637/jss.v045.i03.

Vijayakumar, N., K. L. Mills, A. Alexander-Bloch, C. K. Tamnes, and S. Whittle. 2018. "Structural Brain Development: A Review of Methodological Approaches and Best Practices." *Developmental Cognitive Neuroscience* 33: 129–148. https://doi.org/10.1016/j.dcn.2017.11.008.

Voss, M. W., R. S. Prakash, K. I. Erickson, et al. 2010. "Plasticity of Brain Networks in a Randomized Intervention Trial of Exercise Training in Older Adults." *Frontiers in Aging Neuroscience* 2: 32. https://doi.org/10.3389/fnagi.2010.00032.

Voss, M. W., C. Soto, S. Yoo, M. Sodoma, C. Vivar, and H. van Praag. 2019. "Exercise and Hippocampal Memory Systems." *Trends in Cognitive Sciences* 23, no. 4: 318–333. https://doi.org/10.1016/j.tics.2019.01.006.

White, T. A., S. L. Miller, A. E. Sutherland, B. J. Allison, and E. J. Camm. 2024. "Perinatal Compromise Affects Development, Form, and Function of the Hippocampus Part One; Clinical Studies." *Pediatric Research* 95, no. 7: 1698–1708. https://doi.org/10.1038/s41390-024-03105-7.

Wilckens, K. A., C. M. Stillman, A. M. Waiwood, et al. 2021. "Exercise Interventions Preserve Hippocampal Volume: A Meta-Analysis." *Hippocampus* 31, no. 3: 335–347. https://doi.org/10.1002/hipo.23292.

Wickham, H., M. Averick, J. Bryan, et al. 2019. "Welcome to the Tidyverse." *Journal of Open Source Software* 4, no. 43: 1686. https://doi.org/10.21105/joss.01686.

Wierenga, L. M., M. Langen, B. Oranje, and S. Durston. 2014. "Unique Developmental Trajectories of Cortical Thickness and Surface Area." *Neuroimage* 87: 120–126. https://doi.org/10.1016/j.neuroimage.2013.11.010.

Wu, Y., Y. C. Lu, M. Jacobs, et al. 2020. "Association of Prenatal Maternal Psychological Distress With Fetal Brain Growth, Metabolism, and Cortical Maturation." *JAMA Network Open* 3, no. 1: e1919940. https://doi.org/10.1001/jamanetworkopen.2019.19940.

Yang, Y., M. Lagisz, Y. Z. Foo, D. W. A. Noble, H. Anwer, and S. Nakagawa. 2021. "Beneficial Intergenerational Effects of Exercise on Brain and Cognition: A Multilevel Meta-Analysis of Mean and Variance." *Biological Reviews* 96, no. 4: 1504–1527. https://doi.org/10.1111/brv.12712.

Zhong, S., W. Ding, L. Sun, et al. 2020. "Decoding the Development of the Human Hippocampus." *Nature* 577, no. 7791: 531–536. https://doi.org/10.1038/s41586-019-1917-5.